

Supplementary Information

Interspecies Interactions Drive Community-Level Selection in Microbial Coalescence

Supplementary Methods

Microbial Strain Library, Media, and Culture

We obtained a library of 54 bacterial isolates derived from soil, tree surface, and flower stamen environments collected in Cambridge, MA, USA. Each isolate was purified by serial streaking and stored as glycerol stocks (25% glycerol, -80°C). Phylogenetic identities were determined by full-length 16S rRNA gene sequencing and V4 region amplicon sequencing with BLAST alignment to the SILVA 138 database; the 54 isolates corresponded to 50 unique amplicon sequence variants (ASVs) spanning 29 families across three phyla (Proteobacteria, Firmicutes, and Bacteroidota) (Extended Data Fig. 1; Supplementary Fig. 1). Monoculture growth and pH phenotypes were characterized before community assembly as described below.

The culture medium (CM) for isolates was composed of 1 g L^{-1} yeast extract and 1 g L^{-1} soytone (both from Becton Dickinson), 10 mM sodium phosphate buffer adjusted to pH 6.5, 0.1 mM CaCl_2 , 2 mM MgCl_2 , 4 mg L^{-1} NiSO_4 , and 50 mg L^{-1} MnCl_2 . All media were filter sterilized using bottle-top filtration units (VWR), and all chemicals were purchased from Sigma-Aldrich unless otherwise noted.

Our coalescence experiments were conducted in three media conditions that differed in nutrient supplementation and yielded different failed-invasion frequencies in pairwise assays: (1) Nutr−, CM without additional carbon or nitrogen supplementation; (2) Base, CM with moderate supplementation (5 g L^{-1} glucose and 4 g L^{-1} urea); and (3) Nutr+, CM with high supplementation (20 g L^{-1} glucose and 16 g L^{-1} urea). Both parental community assembly and coalescence experiments were performed in the same media condition throughout each experimental series.

Both monocultures and communities were grown in $300\text{ }\mu\text{L}$ volumes in 96-well deep-well plates (Deepwell plate 96/1000 μL ; Eppendorf) covered with AeraSeal adhesive sealing films (Excel Scientific). Plates were incubated aerobically at 25°C and shaken at 800 r.p.m. on Titramax shakers (Heidolph). To minimize pre-adaptation, all isolates from frozen stocks were preconditioned for two serial growth–dilution cycles prior to assembling parental communities.

Construction of Parental Communities and Community Coalescence

Parental communities were constructed at three inoculated isolate-richness levels: 6, 12, or 24 members (P6, P12, P24), with 30 total communities ($9 \times \text{P6}$, $9 \times \text{P12}$, $12 \times \text{P24}$). For P24 communities, 12 partially overlapping communities were assembled because only two non-overlapping sets of 24 isolates can be drawn from 54 isolates. Coalescence pairs were composed systematically: P6 included 14 pairs, P12 included 27 pairs, and P24 included 6 non-overlapping pairs, resulting in 47 total coalescence pairs for synthetic communities. Some coalescence events were excluded due to sequencing failures. After seven daily growth–dilution cycles, parental communities in Base medium retained a mean ASV richness equivalent to $74 \pm 2\%$ (mean $\pm$ s.e.m.) of their inoculated isolate richness, indicating that Base medium supports high coexistence under the experimental assembly conditions.

Coalescence experiments were conducted by mixing two pre-stabilized parental communities (A and B) at equal volume ratio (1:1) after seven daily dilution cycles. The resulting mixtures were cultured for another seven serial transfers ($\times 30$ dilutions every 24 h) under identical media conditions to obtain post-coalescence endpoint communities (Fig. 1c). Each of the 47 coalescence pairs was performed with two biological replicates, yielding 94 total coalescence events; 11 events were excluded due to sequencing failures or contamination, resulting in 83 coalescence events in Base medium for synthetic community analyses. Additional coalescence experiments were performed in Nutr– and Nutr+ media using extended pair selections; event counts for each condition are reported in the main text (Fig. 4).

16S rRNA Sequencing and Data Processing

Genomic DNA was extracted using QIAGEN DNeasy PowerSoil HTP 96 Kit following the manufacturer’s protocol, with bead-beating for cell lysis. The V4 region of the 16S rRNA gene was amplified using 515F/806R primers containing Illumina overhang adapters and sequenced on an Illumina MiSeq platform (2×250 bp). Raw reads were demultiplexed and denoised using QIIME2 and DADA2¹, generating amplicon sequence variants (ASVs). ASVs were taxonomically classified against the SILVA v138 database².

Relative abundances were normalized by total read depth. ASV tables were filtered to retain only taxa representing $\geq 0.1\%$ relative abundance in any sample. The same numerical cutoff was used as the extinction threshold in simulations for consistency, although sequencing detection thresholds and simulated extinction thresholds have different biological interpretations. The

phylogenetic tree (Supplementary Fig. 1) was constructed using Simple Phylogeny by EMBL’s European Bioinformatics Institute³.

Coalescence Outcome Classification

To characterize coalescence outcomes, we used cosine similarity as the underlying compositional similarity metric between the post-coalescence community and each parental community:

$$\text{Sim}(C, A) = \frac{\vec{x}_C \cdot \vec{x}_A}{\|\vec{x}_C\|_2 \|\vec{x}_A\|_2}, \quad \text{Sim}(C, B) = \frac{\vec{x}_C \cdot \vec{x}_B}{\|\vec{x}_C\|_2 \|\vec{x}_B\|_2}, \quad (1)$$

where $\vec{x}_A$, $\vec{x}_B$, and $\vec{x}_C$ denote community composition vectors, and $\|\cdot\|_2$ denotes the Euclidean norm. Because cosine similarity is invariant to multiplying either vector by a positive scalar, sample-wise total-abundance normalization does not change the similarity score. These cosine-similarity coordinates define the two-parent similarity space, where the position reflects how strongly the coalesced community resembles each parental community.

Because parental communities may share species (overlapping ASVs), classification was based on a linear decomposition of the coalesced community in this normalized two-parent similarity space. We computed similarity-space projection coefficients (u, v) from the weighted combination of parental compositions that reconstructs the coalesced community in this two-parent basis, and then retained only non-negative parental components because negative parental similarity has no biological interpretation. Thus, cosine similarity is the compositional metric, while vector decomposition provides the classification coordinates used to define retention and PDI. When parental communities share no species and the parental composition vectors are L_2 normalized, these coefficients reduce to the cosine similarities. The residual component not represented by either parental community quantifies novel restructuring in the same normalized space.

From these coefficients, we quantified two classification metrics:

1. **Retention magnitude (r):** the coefficient-space retention score from both parental communities, calculated as $r = \sqrt{u^2 + v^2}$. Values close to 1 indicate high similarity-space retention of parental composition, whereas values close to 0 indicate extensive restructuring.
2. **Parental Dominance Index (PDI):** a measure of directional parental similarity, ranging from 0 (complete dominance by parental community B) through 0.5 (equal contributions) to 1 (complete dominance by parental community A). PDI is computed from the angular position in similarity space relative to the diagonal where both parental communities

contribute equally:

$$\text{PDI} = \frac{2}{\pi} \arctan\left(\frac{u}{v}\right) \quad (2)$$

where u and v are the similarity-space projection coefficients for parental communities A and B, respectively. For $v = 0$ and $u > 0$, PDI is defined as the boundary value 1. When $u \gg v$, PDI approaches 1 (parental community A dominance). When $v \gg u$, PDI approaches 0 (parental community B dominance). When $u = v > 0$, PDI equals 0.5 (equal contributions). Events with $u = v = 0$ have zero retention and are classified as Restructuring rather than assigned a directional dominance interpretation.

Based on these metrics, each coalescence event was classified into one of three outcome categories. Restructuring occurs when $r^2 \leq 0.5$, **equivalently when the retention radius satisfies $r \leq 1/\sqrt{2}$ in the two-parental-community similarity map**, meaning that parental-community retention is low in the normalized similarity space, indicating substantial ecological reorganization. Mixture occurs when $r^2 > 0.5$ and PDI is near 0.5 (i.e., $0.25 \leq \text{PDI} \leq 0.75$), indicating high retention with balanced similarity to both parental communities. Dominance occurs when $r^2 > 0.5$ and PDI is near 0 or 1 (i.e., $\text{PDI} < 0.25$ or $\text{PDI} > 0.75$), indicating high retention but with the composition of one parental community dominating that of the other.

We chose the classification thresholds based on two considerations. **First, the retention threshold $r^2 = 0.5$ (radius $r = 1/\sqrt{2}$, not $r = 1/2$) defines a coefficient-space retention boundary in the two-parental-community similarity map.** For the special case of non-overlapping, orthonormal parental composition vectors, this boundary corresponds to half of the squared similarity-space mass lying along the two parental axes. With overlapping parental vectors, it should be interpreted as an operational retention-score boundary rather than a literal explained-composition fraction. Values below this indicate that novel species combinations dominate over parental similarity. Second, the PDI thresholds of 0.25 and 0.75 correspond to contribution ratios of approximately 2.4:1 between parental communities, a level of asymmetry that represents clear dominance of one parental community over the other while allowing for minor similarity to the subordinate parental community. **The robustness of these threshold choices is examined in Supplementary Note 1.**

Lotka–Volterra Simulations

We modeled species-abundance dynamics with the generalized Lotka–Volterra (gLV) system^{4,5}:

$$\frac{dn_i}{dt} = n_i \left(r_i - \sum_{j=1}^S \alpha_{ij} n_j \right) \quad (3)$$

where n_i is the abundance of species i , r_i is its intrinsic growth rate, S is the total number of species, and α_{ij} is the interaction coefficient describing the effect of species j on species i (with self-regulation $\alpha_{ii} = 1$). Under this sign convention, positive α_{ij} values inhibit species i , whereas negative α_{ij} values facilitate species i . Biologically, each α_{ij} is an effective coefficient in the per-capita growth-rate equation. It describes the effect of species j on species i within the coarse-grained gLV model. It can absorb the net inhibitory consequences of direct mechanisms (e.g., resource competition, interference) and indirect mechanisms (e.g., metabolite-mediated inhibition, pH modification) at steady state. It does not identify the underlying biochemical mechanism, and the model does not dynamically represent pH or other environmental state variables. Because the baseline model restricts $\alpha_{ij} \geq 0$, it captures only competitive (growth-inhibiting) interactions. Facilitative interactions such as cross-feeding are not represented in the baseline ensemble. To focus on the interaction-coefficient ensemble, we set all growth rates $r_i = 1$. Off-diagonal interaction coefficients ($i \neq j$) were drawn randomly from a uniform distribution $U[0, 2\mu]$ ^{6,7}. Thus, in the baseline competitive ensemble, μ sets the range of sampled competitive interaction coefficients. This parameter changes the overall scale and heterogeneity of pairwise coefficients, rather than isolating a single biological mechanism. This distribution is a phenomenological ensemble of heterogeneous effective competitive effects rather than a fitted empirical distribution of biochemical mechanisms. The corresponding sensitivity analyses are reported in Supplementary Note 3. Gaussian and Gamma coefficient ensembles preserve the qualitative Mixture-to-Dominance/Restructuring transition, indicating that the result is not specific to the uniform baseline (Supplementary Fig. 6). A separate mean-versus-variance sweep decouples the average inhibitory effect from coefficient heterogeneity and shows that both components contribute to the Dominance regime (Supplementary Fig. 42).

Each baseline simulation used a pool of $N = 54$ species. For each replicate, we randomly permuted the 54 species and assigned the first 48 species sequentially (1–12, 13–24, 25–36, 37–48 after permutation) to form four non-overlapping parental communities of 12 species. The remaining six species were unused in that replicate. This ensured no species were shared across

parental communities, analogous to the experimental design. A fresh interaction matrix was independently sampled for every replicate, so the simulations did not rely on a single fixed species pool.

For single-community assembly, we numerically integrated the gLV equations from random initial conditions $N_i(0) \sim \text{Unif}(0, 0.1)$ to ecological steady state. For the primary simulation sweep, we solved the ODEs with the adaptive Runge–Kutta method RK23 implemented in `scipy.integrate.solve_ivp`, integrating from $t = 0$ to $t = 5000$, which was sufficient for all communities to equilibrate. Performance-optimized robustness simulations used an equivalent vectorized implementation with `scipy.integrate.odeint` while preserving the same integration interval and extinction rule. Following integration, species with relative abundances below an extinction threshold of 10^{-3} were set to zero, yielding four parental-community steady states per replicate.

Community coalescence was simulated by mixing pre-equilibrated parental communities at equal proportions, mirroring the experimental 1:1 volume protocol. For each of the $\binom{4}{2} = 6$ parental community pairs, we retrieved the steady-state community composition vectors $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$, formed the equal mixture $\mathbf{x}_{\text{mix}} = (\mathbf{x}^{(1)} + \mathbf{x}^{(2)})/2$, and zeroed species falling below the extinction threshold in this initial mixture. We then re-integrated the gLV system from $\mathbf{x}_{\text{mix}}$ using the same interaction matrix over $t \in [0, 5000]$ until a new steady state was reached, and reapplied the 10^{-3} threshold to the final abundances. This procedure captures the ecological relaxation after mixing and allows competitive exclusion or coexistence to arise from the underlying interaction structure.

Pairwise Invasion Assays

To operationally estimate invasion resistance across nutrient conditions, we performed pairwise invasion experiments among the 12 most abundant isolates. Each isolate pair was tested in both directions: a 95:5 resident:invader ratio was assembled by colony count and propagated in Nutr−, Base, and Nutr+ media under daily dilution ($\times 30$) for 7 cycles, then the roles were reversed for the reciprocal assay. Final compositions were determined by colony counting on agar plates. Competitive outcomes were classified as: coexistence (both isolates above 10% relative abundance in both directions); exclusion (the same isolate drove its partner below 1% in both directions); or bistability (each isolate excluded the other when acting as resident). Pairs in which the invader established above 1% but below 10% were recorded but not assigned

to a primary outcome class. The fraction of failed invasions was used as an empirical proxy for net interaction intensity and invasion resistance under each nutrient condition⁷⁻⁹. In the gLV calibration this readout corresponds to stronger effective competitive suppression, but it is not interpreted as a direct measurement of a single biochemical mechanism or of all pairwise Lotka-Volterra coefficients.

Natural Sample-Derived Communities

Six environmental samples were collected from diverse microhabitats in Cambridge, MA, USA (soil, compost, decomposing organic matter). Each sample was homogenized and inoculated into culture medium, then subjected to seven serial growth-dilution cycles ($\times 30$ dilution every 24 h) to establish laboratory-stabilized communities. After stabilization, 15 pairwise coalescence events (each with two biological replicates, $n = 30$ per condition) were performed across all three media conditions following the same protocol as synthetic communities. Community composition was characterized via 16S rRNA amplicon sequencing.

Pairwise Selection Correlation Metric

To quantify whether species from the same parental community exhibit correlated selection during coalescence, we developed a pairwise fate-concordance metric. For each coalescence event, species included in the metric were assigned a parental-affiliation label: species detected in only one parental community were assigned to that parent; species detected in both parents were assigned to the parent in which they had higher pre-coalescence relative abundance, with exact ties assigned to parental community A by convention; and species detected in neither parent under the presence threshold were left unassigned and excluded from pair classification. For each species pair, we evaluated whether their presence/absence patterns in the offspring showed shared fates (both present or both absent) or different fates (one survives and the other goes extinct).

The shared-fate rate was converted to a correlation-style concordance metric: $\rho = 2 \times \text{shared-fate rate} - 1$, yielding values from -1 (all pairs have different fates) to $+1$ (all pairs share fate). We computed ρ_{same} for within-parental-affiliation pairs and ρ_{cross} for cross-affiliation pairs, then averaged across coalescence events. To test significance, we generated null distributions by shuffling parental-affiliation labels (1,000 permutations) and computed permutation p -values for $\Delta = \bar{\rho}_{\text{same}} - \bar{\rho}_{\text{cross}}$. This metric is applied to both simulated and experimental coalescence

events in Supplementary Note 7.

Optical Density Measurements and Monoculture Characterization

Optical density (OD) was measured at 600 nm using a plate reader (BioTek Synergy H1) to monitor community growth. Coalescence experiments were initiated by mixing equal volumes of the two stabilized parental communities without normalizing parental OD₆₀₀, so endpoint parental OD₆₀₀ provides an estimate of potential parental-biomass differences at mixing. Post-coalescence samples were measured at each serial dilution cycle to track growth dynamics. All OD measurements were blanked against sterile media controls.

To characterize the phenotypic diversity of isolates prior to community assembly, we measured growth rate and pH modification potential for each of the 54 bacterial isolates in monoculture. OD₆₀₀ was recorded at 24-hour intervals during a 7-day serial dilution regime ($\times 30$ dilution every 24 h) in the culture medium (CM), and growth rate was estimated from the exponential phase. Final pH was measured using a microplate pH meter after 24 h of growth in unbuffered medium to assess each isolate's capacity to acidify or alkalinize the environment. Community-level pH was measured from culture supernatants at the relevant endpoint, rather than continuously during growth. For analyses involving replicated communities or coalescence events, biological replicates were retained as separate event-level observations, matching the treatment used for composition, OD, and outcome analyses. The 54 isolates exhibited broad variation in both growth yield (OD₆₀₀ range: 0.1–1.2) and pH modification potential (final pH range: 4.5–8.5), providing phenotypic context for varied coalescence outcomes (Supplementary Figs. 17, 33).

Statistical Analyses

Most analyses were performed in Python 3.11 using NumPy, pandas, and scikit-learn. Statistical significance was defined at $p < 0.05$ unless otherwise noted. All tests were two-sided unless explicitly stated otherwise.

- Paired t -tests were used to compare simulated interaction-coefficient values before and after community assembly.
- Permutation tests (1,000 permutations) were used to assess whether pairwise selection correlation differed between same-parental-community and cross-parental-community species

pairs, with the null distribution generated by shuffling parental-affiliation labels within each coalescence event (metric defined in “Pairwise Selection Correlation Metric” above).

- Mann-Whitney U tests were used to compare experimental parental-asymmetry values against null model distributions.
- Linear regression was used to quantify the predictability of coalescence outcomes from dominant-species pairwise competition, with R^2 reported as the coefficient of determination.
- χ^2 tests of independence were used to compare observed vs. expected counts of Dominance, Mixture, and Restructuring events; Cochran–Armitage trend tests were used for ordered nutrient-dependent shifts in binary outcome fractions.

Supplementary Note 1: Robustness of coalescence outcome classification and null-model controls

The classification framework, similarity metric, and outcome thresholds used throughout this work are defined in Supplementary Methods. Here we test whether the predominance of Dominance is robust to these analytical choices and whether it could instead arise as an artifact of classification boundaries, compositional metric, abundance skew, or a few dominant species.

Boundary sensitivity of the classification thresholds. To make the boundary dependence explicit, we report the continuous similarity measures underlying the categorical classifications across media in Supplementary Figs. 10–12. A boundary-sensitivity analysis (Supplementary Fig. 37) further shows that the nutrient-dependent increase in Dominance remains evident when either the PDI boundary or retention boundary is varied around the manuscript baseline.

Abundance-skew null models. A potential concern is that Dominance could arise from skewed abundance distributions rather than ecological interactions. To address this, we compared experimental parental-asymmetry values against two null models.

In Null Model 1 (abundance-weighted random selection), species survival probability is proportional to abundance in the combined parental-community pool, testing whether high-abundance species simply survive regardless of parental-community affiliation. In Null Model 2 (shuffled abundance), abundances are randomly permuted among species within each parental

community before neutral mixing, testing whether the overall skewness structure alone drives Dominance.

We generated 500 null offspring communities per model and calculated parental asymmetry, $|2\text{PDI} - 1|$ (Supplementary Fig. 44). Experimental Base-medium data showed mean parental asymmetry of 0.812 ($n = 83$). Both null models produced significantly lower values: abundance-weighted null mean = 0.503 ($U = 32188$, $p = 7.93 \times 10^{-16}$), shuffled null mean = 0.552 ($U = 32050$, $p = 1.85 \times 10^{-15}$; Mann-Whitney U tests). These results indicate that abundance skewness alone cannot explain the observed patterns and support the conclusion that ecological processes beyond simple abundance-weighted survival help determine which parental community dominates.

Simple additive model. We also tested a simple additive null model, $n_{C,\text{null}} = n_A + n_B$, constructed separately for each experimental coalescence event from the raw ASV count vectors of that event’s two parental communities (Extended Data Fig. 3). Under this model, an apparent Dominance classification can arise when the two parental-community raw ASV count vectors have substantially different Euclidean norms: for non-overlapping parental-community raw ASV count vectors, $\text{Sim}(A, A+B)$ and $\text{Sim}(B, A+B)$ are proportional to $\|n_A\|_2$ and $\|n_B\|_2$, respectively. This norm-imbalance effect is possible in principle, especially when richness is low and vector norms are sensitive to abundance unevenness among a small number of taxa. In the experimental data, however, the simple additive null model was usually classified as Mixture, including 66/83 Base medium events (79.5%), and the observed Base medium outcomes were shifted further toward parental asymmetry than their corresponding null-model outcomes (one-sided paired Wilcoxon test, $p = 5.9 \times 10^{-14}$). Thus, the simple additive null model does not explain the observed Dominance tendency. Related geometric effects can arise more generally when parental-community composition vectors differ in norm or unevenness; these are evaluated both with the simple additive null model above and with the simulation parental-community composition-vector norm analysis in Supplementary Fig. 38. Together, these analyses show that such geometric effects can produce apparent Dominance in principle, but do not account for most experimental coalescence outcomes or the simulated Dominance transition.

Sensitivity of outcome classification to the compositional metric and its biological interpretation. A further concern is whether the predominance of Dominance is an artifact of the particular compositional metric used to define the outcome categories rather than a ro-

bust feature of the data. To test this, we compared coalescence outcome classifications using five approaches: the primary cosine-similarity/vector-decomposition framework, Euclidean distance, Bray-Curtis dissimilarity, Jensen-Shannon divergence, and Jaccard index. The alternative relative-abundance approaches, Euclidean distance and Bray-Curtis dissimilarity, produced outcome distributions broadly consistent with the primary framework, with Dominance as the most frequent outcome. However, Jensen-Shannon divergence and Jaccard index did not recover this pattern (Extended Data Fig. 2), reflecting the sensitivity of boundary assignments to the choice of compositional metric. We interpret this divergence as biologically informative rather than solely technical: the primary framework, Euclidean distance, and Bray-Curtis retain quantitative abundance information, whereas Jaccard collapses the data to presence/absence and therefore measures species-identity retention rather than abundance-weighted parental dominance. The Dominance pattern is therefore robust across abundance-weighted parental-similarity metrics. Its absence under Jaccard indicates that Dominance is not primarily a presence/absence retention signal, while the Jensen-Shannon result indicates that full-distribution divergence emphasizes a different aspect of community composition than directional parental similarity.

Effect of dominant-species abundance on PDI and the Dominance trend. Because PDI is abundance-weighted, a single highly abundant taxon shared between the coalesced community and one parental community can dominate the composition vectors and inflate the directional similarity toward that parent. This raises the concern that the nutrient-dependent increase in Dominance (Fig. 4D) and the Fig. 5C predictability analysis could partly reflect a mechanical contribution from a few dominant species rather than a community-level reorganization toward one parental community—that is, the association could be partly circular, because the same dominant taxa that define the coalesced community also determine its PDI. To test this, we recalculated PDI after removing either the coalesced-community dominant species or the two parental-community dominant species from all three composition vectors, so that the metric reflects only the subdominant community structure. The Nutr+ association weakened but remained significant, whereas Base was not significant after removal (Supplementary Fig. 16). Thus, the Nutr+ Dominance signal is partly, but not entirely, attributable to the most abundant species; a community-wide contribution beyond the dominant taxa remains.

Supplementary Note 2: Shared assembly history promotes Dominance

To test whether assembly history contributes to community-level selection¹⁰⁻¹³, we compared coalescence outcomes to a “direct assembly” comparison. In coalescence, two parental communities are first assembled separately from the species pool and then mixed. In direct assembly, all species from both pools are mixed simultaneously without prior assembly history.

In simulations, species compete during assembly and some go extinct. Those that survive to coexistence tend to have weaker mutual interactions on average. We quantified this assembly effect by measuring the average pairwise interaction coefficient $\langle\alpha_{ij}\rangle$ among species at different stages (Supplementary Fig. 32) and by visualizing pre- and post-assembly interaction submatrices (Supplementary Fig. 3). Across all tested interaction-strength parameter values ($\mu = 0$ to 1.2), post-assembly communities exhibited substantially reduced average pairwise interaction coefficients compared to the pre-assembly pool; at the representative value $\mu = 0.6$, this reduction was significant across 600 paired pre- versus post-assembly summaries (paired t -test, $t_{599} = 29.26$, $p = 1.54 \times 10^{-117}$; Fig. 2C). This reduction becomes more pronounced at higher μ , where stronger competition drives more species to extinction, leaving only those with weaker mutual interactions. The assembly process thus creates internal coherence within each parental community: species that survived together share compatible (less competitive) interactions.

We hypothesized that this shared history would increase Dominance probability compared to direct assembly. Simulations supported this prediction (Extended Data Fig. 8): across interaction-strength parameter values $\mu = 0.2$ to 1.0, coalescence consistently produced higher Dominance fractions than direct assembly. At $\mu = 0.2$, coalescence yielded 37% Dominance versus 12% for direct assembly. At $\mu = 0.6$, the values were 59% versus 20%. At $\mu = 1.0$, they were 65% versus 22%. Correspondingly, direct assembly showed higher Restructuring rates, as expected when species from different pools have not been pre-filtered for compatibility.

Experimental results corroborate these findings (Supplementary Fig. 23). Under Base and Nutr+ conditions, coalescence showed elevated Dominance rates compared to direct assembly, consistent with simulation predictions. The Nutr- condition showed a weaker coalescence-versus-direct-assembly contrast, consistent with weaker invasion resistance in nutrient-limited environments (see Supplementary Note 4). A separate pair-level analysis of parental-affiliation-correlated species fates is presented in Supplementary Note 7.

Supplementary Note 3: Robustness of the simulated Dominance transition to model assumptions

The baseline gLV model and the interaction-strength parameter μ are defined in Supplementary Methods. Here, μ provides a minimal one-parameter control of effective interaction strength that jointly sets the average inhibitory effect and the heterogeneity of pairwise coefficients^{5–7,9,14,15}. We test whether the simulated transition from Mixture-dominated to Dominance/Restructuring outcomes with increasing μ is robust to the assumptions of that baseline, including the coefficient distribution, demographic heterogeneity in growth rates and carrying capacities, community size, interaction sign structure, and the decomposition of μ into coefficient mean and variance.

Baseline gLV sensitivity analyses

We first tested whether heterogeneity in intrinsic growth rates or carrying capacities altered the qualitative transition. Allowing species-level variation in growth rates or carrying capacities changed the detailed outcome frequencies but did not remove the increase in Dominance with the interaction-strength parameter μ (Supplementary Figs. 4–5).

We next tested whether the qualitative transition depended on the shape of the coefficient distribution. We compared three distributions with matched coefficient mean μ : uniform ranging from 0 to 2μ (primary analysis), Gaussian with mean μ and standard deviation $\mu/\sqrt{3}$, and Gamma with mean μ and shape parameter 3. All three distributions were parameterized to have the same mean μ and matched variance $\mu^2/3$ (for the uniform distribution $U[0, 2\mu]$, $\text{var} = (2\mu)^2/12 = \mu^2/3$; for the Gaussian, exact; for the Gamma, exact). The qualitative transition from Mixture-dominated to Dominance/Restructuring outcomes with increasing interaction-strength parameter μ persisted across all tested distributions (Supplementary Fig. 6).

We also tested whether community size (number of species per parental community) affects the frequency of Dominance outcomes. Simulations were performed with parental community sizes ranging from 4 to 48 species per community (Extended Data Fig. 4). In this ablation, the total simulated species pool scaled with parental-community richness as four non-overlapping parental communities ($N = 16, 24, 36, 48, 96$, or 192 species for 4, 6, 9, 12, 24, or 48 species per community, respectively). Dominance frequency changed little across community sizes (~ 14 – 26% at $\mu = 0.3$, ~ 56 – 67% at $\mu = 0.6$, ~ 69 – 78% at $\mu = 0.8$), while Mixture decreased and Restructuring increased with larger species pools. This pattern reflects increased opportunity for competitive exclusion cascades in larger communities, but Dominance remained common at

moderate-to-high interaction-strength parameter values.

Beneficial-interaction and reciprocal pair-coupling extensions

Because our baseline gLV ensemble is restricted to non-negative, growth-inhibiting coefficients, we also tested additional interaction structures while applying the same assembly, coalescence, and outcome-classification pipeline. First, we introduced a mixed-sign facilitative-tail model in which off-diagonal coefficients were sampled from $\alpha_{ij} \sim U[-f\mu, (2+f)\mu]$, preserving $E[\alpha_{ij}] = \mu$ while allowing a small fraction of coefficients to be negative under our sign convention (positive ecological interactions). To prevent runaway growth in the presence of facilitation, we added density-dependent self-limitation, $\dot{n}_i = n_i(1 - (An)_i - 0.1n_i^2)$. Across $f \in \{0, 0.10, 0.20, 0.40, 0.80\}$ and $\mu \in \{0.30, 0.60, 0.80\}$, Dominance still increased with μ , although increasing the facilitative tail shifted some low- μ outcomes from Mixture toward Dominance and Restructuring (Supplementary Fig. 39).

Second, we varied the reciprocal pair-coupling structure between coefficients α_{ij} and α_{ji} . For a fraction $|p|$ of unordered species pairs, coefficients were constrained to follow an antisymmetric exploitation structure when $p < 0$ ($\alpha_{ji} = -\alpha_{ij}$) or a symmetric competition structure when $p > 0$ ($\alpha_{ji} = \alpha_{ij}$). Remaining pairs were drawn iid from the baseline competitive distribution. This pair-coupling sweep tests whether the Dominance increase depends on independent directed interaction coefficients. The qualitative μ -dependent increase in Dominance persisted across the reciprocal pair-coupling sweep and in a finer sweep from $p = -1$ to $p = +1$ (Supplementary Fig. 40).

Third, to address the specific mutualism question, we converted a controlled fraction of unordered species pairs into weak bidirectional mutualisms. For $q \in \{0, 0.10, 0.20, 0.30, 0.40\}$ of species pairs, both reciprocal coefficients were sampled from $\alpha_{ij}, \alpha_{ji} \sim U[-0.2\mu, 0]$, while all remaining off-diagonal coefficients were sampled from the baseline competitive distribution $U[0, 2\mu]$. We used the same density-dependent self-limitation and the same assembly, coalescence, and classification pipeline as in the mixed-sign facilitative-tail analysis. Weak mutualistic pairs did not eliminate the interaction-strength-dependent Dominance trend: at $\mu = 0.60$, Dominance was 58.5%, 70.2%, 69.8%, 67.3%, and 58.7% for $q = 0, 0.10, 0.20, 0.30, 0.40$, respectively, and at $\mu = 0.80$ it remained 69.4–77.5% across the same sweep (Supplementary Fig. 41).

Mean and heterogeneity of interaction coefficients

Fourth, to test which components of this one-parameter interaction-strength axis are required, we separated the mean and variance of the interaction-coefficient distribution. Off-diagonal coefficients were sampled from $\alpha_{ij} \sim U[m - h, m + h]$ for $m \in \{0, 0.2, 0.4, 0.6, 0.8\}$ and $h \in \{0, 0.2, 0.4, 0.6, 0.8\}$, giving $\text{std}(\alpha_{ij}) = h/\sqrt{3}$. Negative coefficients, corresponding to positive ecological interactions under our sign convention, occur when $h > m$. When $h = 0$, all outcomes were Mixture even at $m = 0.8$, showing that coefficient mean alone is not sufficient to generate Dominance. Increasing h at fixed m increased Dominance and same-parent fate concordance. The strongest Dominance values occurred when both the coefficient mean and coefficient variance were high. Across the 25 cells, Dominance correlated more strongly with h (and therefore coefficient standard deviation) than with m (descriptive Pearson $r = 0.78$ versus 0.53 ; Supplementary Fig. 42). Thus, both average inhibitory effect and coefficient heterogeneity contribute to the effective interaction-strength axis captured by the baseline μ parameter. These sensitivity checks indicate that the main simulation result is not restricted to iid non-negative coefficients. Strongly mutualistic or facilitation-dominated systems remain outside the baseline model scope^{12,16}.

Supplementary Note 4: Nutrient dependence of pairwise invasion resistance

To operationally estimate pairwise invasion resistance, we performed reciprocal invasion assays using the 12 most abundant isolates (ranked by their corresponding ASV abundances across the sequencing dataset), testing each pair in both directions to assess competitive outcomes and detect potential bistability. Full assay protocol (95:5 resident:invader ratio, 7-cycle propagation, colony counting, and classification thresholds) is described in Supplementary Methods.

Invasion assays were performed across all three nutrient conditions: Nutr−, Base, and Nutr+. We use failed invasion as the operational assay readout for competitive exclusion, defined as cases in which the introduced species remained below 1% relative abundance after propagation. Among the assayed abundant isolates, the frequency of failed invasion increased with nutrient concentration, consistent with the coalescence outcome patterns observed in community experiments (Supplementary Figs. 7–9). Under Nutr−, more pairs achieved stable coexistence, while Nutr+ conditions led to more frequent exclusion of one species by the other. Higher failed-invasion frequency among these assayed isolates is consistent with stronger resistance to cross-community establishment^{7,8} and with prior work linking nutrient and interaction strength

to microbial competitive dynamics^{6,9}, which can generate more correlated parental-community fates after coalescence. This provides a plausible link by which pairwise invasion resistance could favor Dominance rather than balanced Mixture, although these assays alone do not establish causality beyond the evidence shown in the figures.

Supplementary Note 5: Alternative explanations for the nutrient-dependent Dominance

Unequal parental biomass at mixing

Several alternative mechanisms could generate parental-affiliation-correlated persistence during coalescence^{17,18}, including passive effects of unequal parental biomass at mixing, initial-richness effects, and environmental modification by dominant taxa. Because coalescence was initiated by equal-volume pooling without OD normalization, parental-community biomass differences at mixing could bias the coalesced community toward the higher-biomass parent. To test this hypothesis, we asked whether parental-community OD₆₀₀ correlates with parental dominance direction or with pairwise selection correlation. If Dominance were driven primarily by unequal absolute biomass at mixing, the higher-OD parental community should win more frequently. Signed ΔOD_{A-B} , the difference between the OD of parental community A and parental community B, should also be positively associated with PDI. We did not observe these predictions. The higher-OD parental community won only 13/35 Nutr− events (37.1%; two-sided binomial test, $p = 0.18$), 20/54 Base events (37.0%; $p = 0.076$), and 9/68 Nutr+ events (13.2%; $p = 3.9 \times 10^{-10}$) among Dominance outcomes (Supplementary Fig. 45). Signed ΔOD_{A-B} was not positively associated with PDI in any medium: the Spearman rank correlations were negative in Nutr− ($n = 90$, $\rho = -0.24$, $p = 0.021$), weak and not statistically significant in Base ($n = 83$, $\rho = -0.14$, $p = 0.19$), and strongly negative in Nutr+ ($n = 90$, $\rho = -0.60$, $p = 4.0 \times 10^{-10}$; Supplementary Fig. 14). OD-binned within-community pairwise selection correlation did not increase consistently across media (Supplementary Fig. 15). The negative association in Nutr+ (lower-OD communities winning more often) is consistent with the pH-mediated mechanism explored below: acidifying parental communities may yield lower endpoint biomass under high-nutrient conditions yet still win coalescence through environmental modification. Thus, parental-community biomass differences do not fully explain the nutrient-dependent increase in Dominance or correlated species fates.

Initial richness and pool size

We also tested whether initial community richness could account for the observed community-level selection signatures in the experiments and simulations (Supplementary Fig. 13). In the synthetic-community experiments, we did not detect a statistically significant effect of inoculated isolate richness on the full three-category outcome distribution across 6, 12, and 24 inoculated isolates ($\chi^2 = 2.24$, $p = 0.69$). We also did not detect a pool-size effect within individual nutrient conditions (Dominance vs non-Dominance tests: Nutr- $p = 0.39$, Base $p = 0.82$, Nutr+ $p = 0.27$). Initial inoculated richness affected assembly structure: realized parental ASV richness increased with pool size, whereas ASV richness per inoculated isolate decreased with pool size. These assembly-richness effects did not produce a corresponding pool-size dependence of Dominance frequency.

In the gLV model, we performed a pool-size ablation at $\mu \in \{0.30, 0.60, 0.80\}$, varying the initial richness from 4 to 48 species per parental community while scaling the total pool as four non-overlapping parental communities ($N = 16$ –192 species). Dominance frequency was primarily set by μ and changed little with pool size, whereas the same-parental-community versus cross-parental-community pairwise selection correlation gap increased with pool size (Supplementary Fig. 13). This supports the interpretation that pool size affects assembly filtering and correlated species fates more than it affects Dominance-class frequency. Thus, pool-size variation does not fully explain the main transition in community-level selection; instead, it modulates the strength of correlated species fates within the interaction-strength regimes described in the main text.

pH-mediated environmental modification

We considered environmental-feedback models motivated by pH-mediated microbial interactions^{9,19}. A Ratzke–Gore-style pH-feedback model can generate asymmetric replacement over an intermediate feedback range, showing that environmental modification alone can contribute to Dominance-like outcomes (Supplementary Fig. 43). However, in our implementation this pH-only model did not reproduce the monotonic increase in Dominance with the interaction-strength parameter seen in the effective-interaction gLV framework and produced rare Restructuring outcomes ($\sim 1\%$; 3/297 simulated events), in contrast to the higher Restructuring fraction in the effective-interaction gLV framework ($\sim 22\%$ across μ values). This result and the strict empirical pH-contrast analysis (Extended Data Fig. 7) support pH modification as a plausible contributor

to winner identity in Nutr+, rather than as a complete explanation for the observed Nutr+ Dominance frequency or as a replacement for the outcome-level community-level selection interpretation.

In the strict acidic–alkaline subset, where one parental community had $\text{pH} < 6.5$ and the other had $\text{pH} > 7.5$, acidic parental communities dominated 29/32 Nutr+ events (90.6%), compared with 23/41 Base events (56.1%; Extended Data Fig. 7; Supplementary Fig. 46). A separate frequency comparison showed that strict acidic–alkaline pairings were enriched for Dominance relative to strict same-pH pairings in Nutr+ (29/32 versus 22/34 events; Fisher $p = 0.018$), but not in Base medium (24/41 versus 23/32 events; Fisher $p = 0.33$; Supplementary Fig. 46). This strict high-contrast analysis supports pH-mediated winner-direction bias in Nutr+, but it does not by itself explain the full Dominance pattern across all coalescence events because same-pH Nutr+ pairings often produced Dominance and intermediate-pH pairings were excluded.

Supplementary Note 6: Dominance extends to laboratory-stabilized natural communities, with limits on generality

The natural sample-derived community experiments test coalescence after laboratory stabilization, not coalescence of unfiltered environmental communities. Each environmental inoculum underwent seven serial growth-dilution cycles in defined laboratory media before coalescence. This enrichment phase may pre-select taxa and metabolic strategies that thrive under the culture conditions²⁰, potentially reducing effective ecological heterogeneity and making natural sample-derived communities more similar to the synthetic consortia.

The available post-stabilization 16S data indicate that this filtering was not complete at the ASV level. Natural sample-derived parental communities retained higher richness than the synthetic parental communities (13.7 ± 7.2 versus 9.8 ± 4.8 ASVs above the 0.1% relative-abundance threshold; Supplementary Figs. 19–21). Stabilized parental replicates from the same environmental source sample were more similar to one another than parental communities from different source samples at both ASV and genus levels. Coalesced communities also remained closer to their own parental-community pair than to unrelated same-medium parental communities (Supplementary Fig. 22). These analyses argue against complete ASV- or genus-level taxonomic convergence to a single shared endpoint, but they do not quantify how much convergence occurred during stabilization.

Because the original environmental inocula were not sequenced before enrichment, we can-

not measure pre-to-post taxonomic convergence during the stabilization phase. We also cannot directly test functional convergence because these experiments used 16S community profiling rather than metagenomic, metabolomic, or trait-resolved measurements. Thus, pre-selection could contribute to the convergent coalescence patterns observed between synthetic and natural sample-derived communities, and the natural-community results should be interpreted as evidence that the nutrient-dependent increase in Dominance occurs in taxonomically richer laboratory-stabilized communities, not as evidence for unrestricted generality across unfiltered natural ecosystems^{18,21–23}. Future work using culture-independent time series or broader environmental sampling could help disentangle laboratory adaptation from intrinsic ecological dynamics.

Supplementary Note 7: Interpretation of pairwise selection correlation

Metric interpretation. The pairwise selection-correlation metric is defined in Supplementary Methods. Positive $\bar{\rho}_{\text{same}}$ indicates that species with the same parental-affiliation label share fates more often than they have different fates, whereas negative $\bar{\rho}_{\text{cross}}$ indicates that cross-affiliation species pairs have different fates more often than shared fates. Thus, $\Delta = \bar{\rho}_{\text{same}} - \bar{\rho}_{\text{cross}}$ measures how strongly survival or extinction becomes correlated within assembled parental communities relative to pairs drawn from different parental communities.

Simulation evidence. In simulations (Extended Data Fig. 5a–d), same-parent pairs remained positively correlated, whereas cross-community pairs became increasingly negative as the interaction-strength parameter increased. At low interaction-strength parameter values ($\mu = 0.3$), both within-community and cross-community pairs show positive correlations with only a small difference ($\Delta = 0.05$; two-sided permutation test, $p < 0.001$; Extended Data Fig. 5a). As the interaction-strength parameter increases, the difference becomes pronounced. At $\mu = 0.6$, within-community pairs show positive correlation while cross-community pairs become negative ($\Delta = 0.59$, $p < 0.001$; Extended Data Fig. 5b). At $\mu = 0.8$, the separation is larger ($\Delta = 0.94$, $p < 0.001$; Extended Data Fig. 5c).

Experimental evidence. In experiments (Extended Data Fig. 6), the same-versus-cross separation was statistically significant in Base ($n = 83$, $\Delta = 0.235$, $p < 0.001$) and Nutr+ ($n = 83$, $\Delta = 0.141$, $p < 0.001$), but not in Nutr− ($n = 90$, $\Delta = 0.016$, $p = 0.307$). The separation was largest in Base, and the significant Nutr+ separation indicates parental-affiliation-correlated

species fates under a condition with higher failed-invasion frequency and stronger environmental feedbacks.

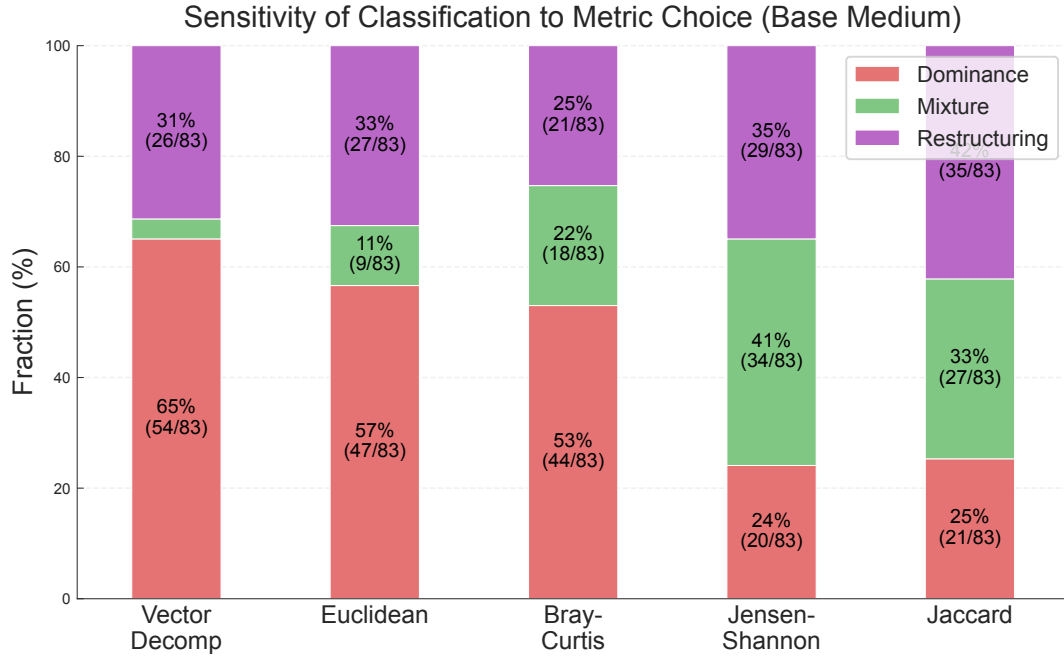
Biological interpretation. Together, these patterns indicate that assembly history structures species-fate concordance during coalescence: species from the same assembled parental community tend to persist or be lost together, whereas species from different parental communities become increasingly anti-concordant as interactions strengthen. This provides species-fate-level support for interpreting Dominance outcomes as community-level selection [12,13,17,18](#).

Relationship to invasion fitness. This metric can also be interpreted through the invasion-fitness framework. Related pairwise invasion-resistance assays are summarized in Supplementary Note 4. In a gLV model, invasion fitness is the per-capita growth rate of a rare species introduced into a resident community background. Positive invasion fitness predicts establishment, whereas negative invasion fitness predicts exclusion [7,8](#). The 95:5 pairwise invasion assays provide an empirical two-species approximation of this idea, while the pairwise fate-concordance metric asks whether analogous persistence outcomes become correlated across many species from the same assembled parental community during coalescence. Consistent with this interpretation, an auxiliary gLV analysis showed that excess same-parent shared fate, measured relative to a parental-affiliation-shuffled null, increased with the interaction-strength parameter μ across the full simulated sweep ($\mu = 0.05\text{--}1.20$; Pearson $r = 0.870$, $p = 3.2 \times 10^{-8}$; Supplementary Fig. 36). This auxiliary within-parent metric measures the same parental-affiliation-correlated shared-fate signal as $\Delta = \bar{\rho}_{\text{same}} - \bar{\rho}_{\text{cross}}$, but compares same-parent pairs to a parental-affiliation-shuffled null rather than explicitly subtracting cross-community pairs.

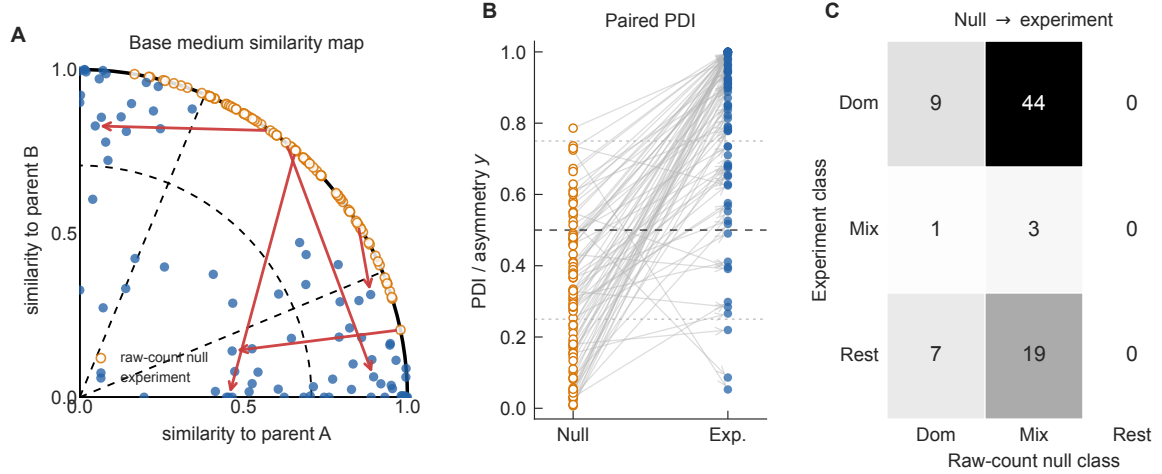
Extended Data Figures

ASV1	k-Bacteria, p-Firmicutes, c-Bacilli, o-Lactobacillales, f-Streptococcaceae, g-Lactococcus
ASV2	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Enterobacterales, f-Enterobacteriaceae, g-Raoultella
ASV3	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Enterobacterales, f-Enterobacteriaceae, g-Klebsiella
ASV4	k-Bacteria, p-Bacteroidota, c-Bacteroidia, o-Flavobacteriales, f-Weeksellaceae, g-Chryseobacterium
ASV5	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Enterobacterales, f-Enterobacteriaceae, g-Pluralibacter
ASV6	k-Bacteria, p-Firmicutes, c-Bacilli, o-Lactobacillales, f-Leuconostocaceae, g-Leuconostoc
ASV7	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Aeromonadales, f-Aeromonadaceae, g-Aeromonas
ASV8	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Enterobacterales, f-NA, g-NA
ASV9	k-Bacteria, p-Bacteroidota, c-Bacteroidia, o-Flavobacteriales, f-Weeksellaceae, g-Empedobacter
ASV10	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Xanthomonadales, f-Xanthomonadaceae, g-Stenotrophomonas
ASV11	k-Bacteria, p-Bacteroidota, c-Bacteroidia, o-Sphingobacteriales, f-Sphingobacteriaceae, g-Sphingobacterium
ASV12	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Enterobacterales, f-Erwinaceae, g-Pantoea
ASV13	k-Bacteria, p-Firmicutes, c-Bacilli, o-Lactobacillales, f-Leuconostocaceae, g-Leuconostoc
ASV14	k-Bacteria, p-Proteobacteria, c-Alphaproteobacteria, o-Rhizobiales, f-Rhizobiaceae, g-Ochrobactrum
ASV15	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Pseudomonadales, f-Pseudomonadaceae, g-Pseudomonas
ASV16	k-Bacteria, p-Firmicutes, c-Bacilli, o-Exiguobacterales, f-Exiguobacteraceae, g-Exiguobacterium
ASV17	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Enterobacterales, f-Enterobacteriaceae, g-Escherichia/Shigella
ASV18	k-Bacteria, p-Firmicutes, c-Bacilli, o-Bacillales, f-Planococcaceae, g-Lysinibacillus
ASV19	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Pseudomonadales, f-Moraxellaceae, g-Acinetobacter
ASV20	k-Bacteria, p-Bacteroidota, c-Bacteroidia, o-Flavobacteriales, f-Weeksellaceae, g-Empedobacter
ASV21	k-Bacteria, p-Bacteroidota, c-Bacteroidia, o-Flavobacteriales, f-Weeksellaceae, g-Empedobacter
ASV22	k-Bacteria, p-Firmicutes, c-Bacilli, o-Staphylococcales, f-Staphylococcaceae, g-Staphylococcus
ASV23	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Pseudomonadales, f-Pseudomonadaceae, g-Pseudomonas
ASV24	k-Bacteria, p-Bacteroidota, c-Bacteroidia, o-Sphingobacteriales, f-Sphingobacteriaceae, g-Pedobacter
ASV25	k-Bacteria, p-Bacteroidota, c-Bacteroidia, o-Cytophagales, f-Spirosomaceae, g-Flectobacillus
ASV26	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Burkholderiales, f-Oxalobacteraceae, g-Herbaspirillum
ASV27	k-Bacteria, p-Firmicutes, c-Bacilli, o-Bacillales, f-Bacillaceae, g-Bacillus
ASV28	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Pseudomonadales, f-Pseudomonadaceae, g-Pseudomonas
ASV29	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Xanthomonadales, f-Xanthomonadaceae, g-Stenotrophomonas
ASV30	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Burkholderiales, f-Oxalobacteraceae, g-Undibacterium
ASV31	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Enterobacterales, f-Enterobacteriaceae, g-NA
ASV32	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Enterobacterales, f-Enterobacteriaceae, g-Raoultella
ASV33	k-Bacteria, p-Bacteroidota, c-Bacteroidia, o-Cytophagales, f-Spirosomaceae, g-Flectobacillus
ASV34	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Burkholderiales, f-Comamonadaceae, g-Acidovorax
ASV35	k-Bacteria, p-Bacteroidota, c-Bacteroidia, o-Flavobacteriales, f-Weeksellaceae, g-Chryseobacterium
ASV36	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Enterobacterales, f-Enterobacteriaceae, g-Citrobacter
ASV37	k-Bacteria, p-Firmicutes, c-Bacilli, o-Bacillales, f-Planococcaceae, g-NA
ASV38	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Enterobacterales, f-Erwinaceae, g-Pantoea
ASV39	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Pseudomonadales, f-Pseudomonadaceae, g-Pseudomonas
ASV40	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Enterobacterales, f-Enterobacteriaceae, g-Klebsiella
ASV41	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Enterobacterales, f-Enterobacteriaceae, g-Enterobacter
ASV42	k-Bacteria, p-Firmicutes, c-Bacilli, o-Bacillales, f-Bacillaceae, g-Bacillus
ASV43	k-Bacteria, p-Firmicutes, c-Bacilli, o-Exiguobacterales, f-Exiguobacteraceae, g-Exiguobacterium
ASV44	k-Bacteria, p-Bacteroidota, c-Bacteroidia, o-Flavobacteriales, f-Flavobacteriaceae, g-Flavobacterium
ASV45	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Pseudomonadales, f-Pseudomonadaceae, g-Pseudomonas
ASV46	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Enterobacterales, f-NA, g-NA
ASV47	k-Bacteria, p-Actinobacteriota, c-Actinobacteria, o-Streptomycetales, f-Streptomycetaceae, g-Streptomyces
ASV48	k-Bacteria, p-Actinobacteriota, c-Actinobacteria, o-Micrococcales, f-Microbacteriaceae, g-Curtobacterium
ASV49	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Pseudomonadales, f-Pseudomonadaceae, g-Pseudomonas
ASV50	k-Bacteria, p-Proteobacteria, c-Gammaproteobacteria, o-Pseudomonadales, f-Moraxellaceae, g-Acinetobacter

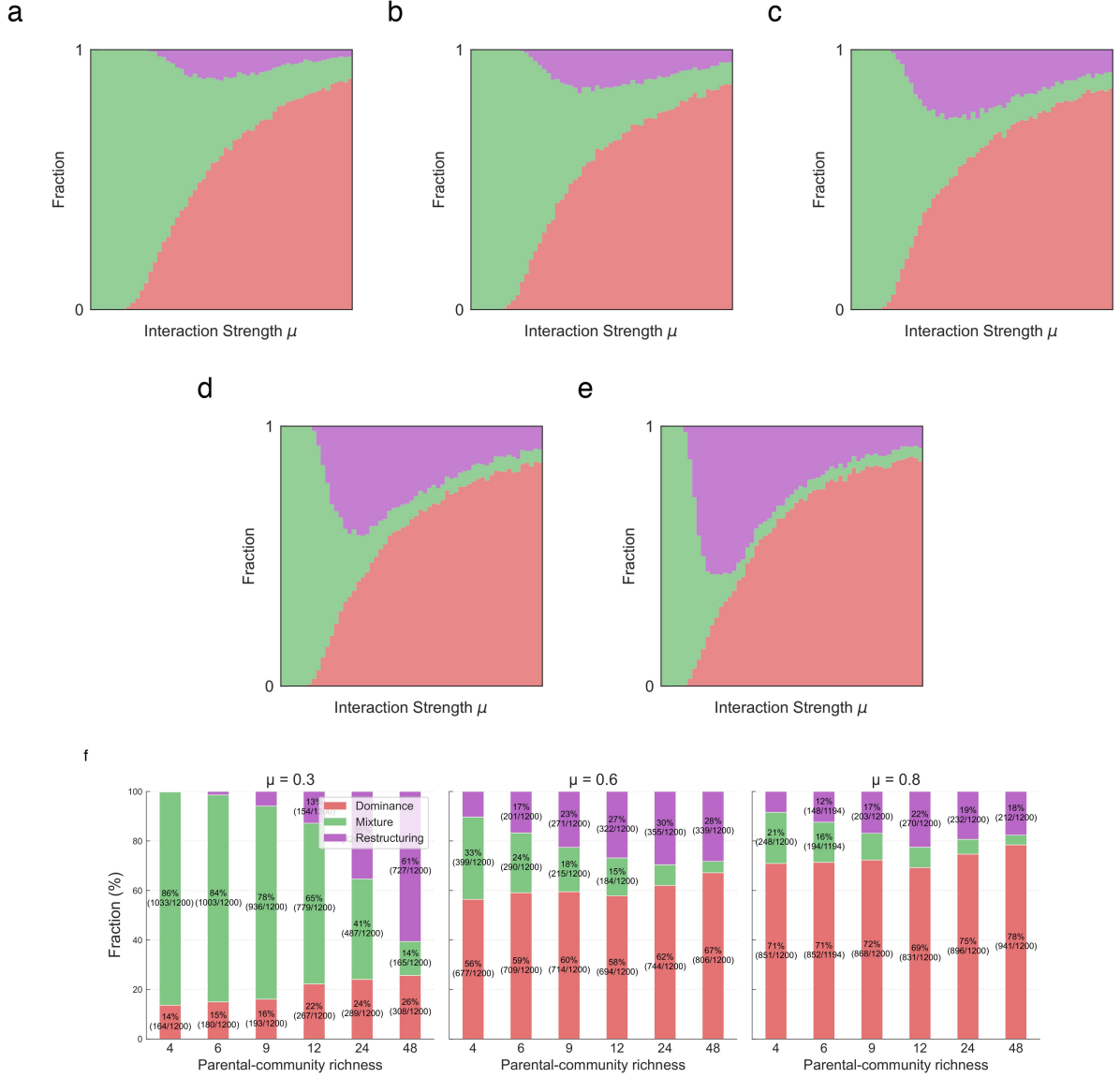
Extended Data Fig. 1. Taxonomic classification of bacterial isolates. Table showing the 54 bacterial isolates used in the experiment, of which 4 pairs share identical ASV sequences. Each row shows the ASV identifier and its full taxonomic classification (Kingdom, Phylum, Class, Order, Family, Genus). This isolate set spans broad taxonomic diversity used to construct the synthetic communities in the coalescence experiments. Colors match those used in pie charts and composition figures throughout the manuscript.



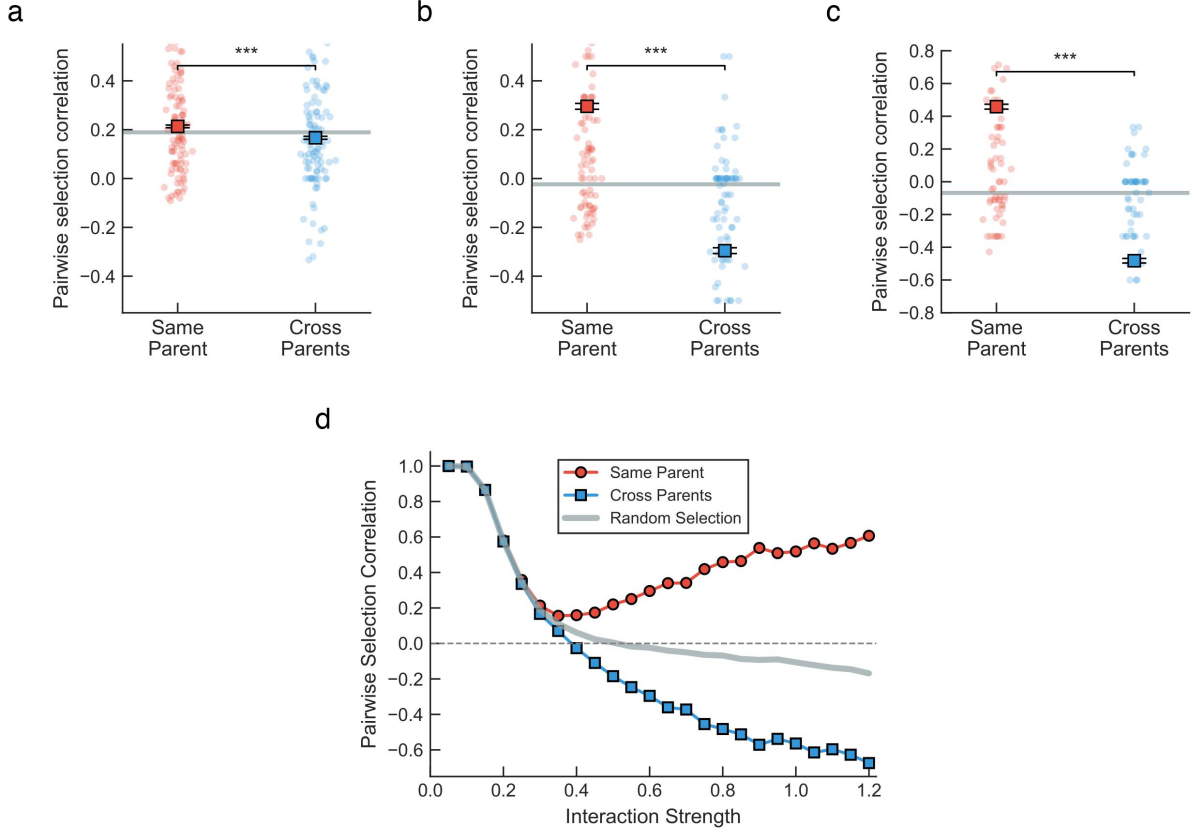
Extended Data Fig. 2. Sensitivity of outcome classification to metric choice. Stacked bar plot showing the distribution of coalescence outcomes (Dominance, Mixture, Restructuring) in Base medium using the primary similarity-map classification and four alternative metrics: Euclidean distance, Bray–Curtis dissimilarity, Jensen–Shannon divergence, and Jaccard index. Percentages and counts (n/total) are shown for each category. **Dominance is recovered by abundance-weighted parental-similarity metrics, including the primary similarity-map classification, Euclidean distance, and Bray–Curtis dissimilarity, but not by Jaccard index or Jensen–Shannon divergence.** This divergence reflects the different biological questions asked by each metric: abundance-weighted metrics quantify whether the coalesced community quantitatively resembles one parent more than the other, whereas Jaccard index measures presence/absence retention and Jensen–Shannon divergence measures differences across the full abundance distribution. Thus, Jaccard and Jensen–Shannon are best interpreted as complementary metric choices that emphasize different features of community composition.



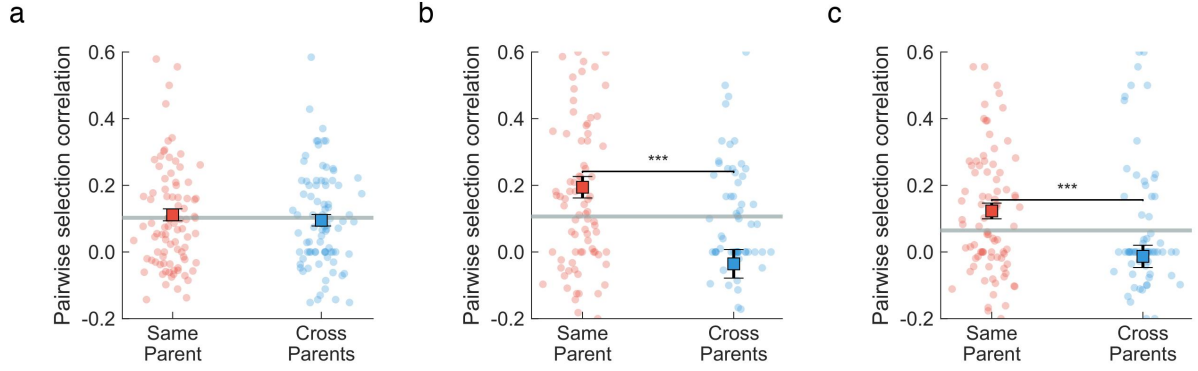
Extended Data Fig. 3. Simple additive model for Base medium coalescence outcomes. This event-matched null model tests whether Dominance can arise from a simple additive null process or parental-community norm imbalance rather than correlated competitive selection. **a**, Similarity map comparing simple additive null-model outcomes (orange open points; $n_{C, \text{null}} = n_A + n_B$, constructed by summing each coalescence pair's two parental-community raw ASV count vectors before cosine/vector-decomposition normalization) with observed coalesced communities (blue filled points); arrows show example null-to-observed pairs. **b**, Paired shifts in direction-independent parental asymmetry, $|2\text{PDI} - 1|$, from each simple additive null-model outcome to the corresponding observed coalesced community for all 83 Base medium events. **c**, Class transitions from simple additive null-model outcomes (columns) to observed coalesced communities (rows). The simple additive null model is usually classified as Mixture, and the observed Base medium outcomes are shifted further toward parental asymmetry than their corresponding null model outcomes (one-sided paired Wilcoxon test, $p = 5.9 \times 10^{-14}$). See Supplementary Note 1 for full analysis and additional null models.



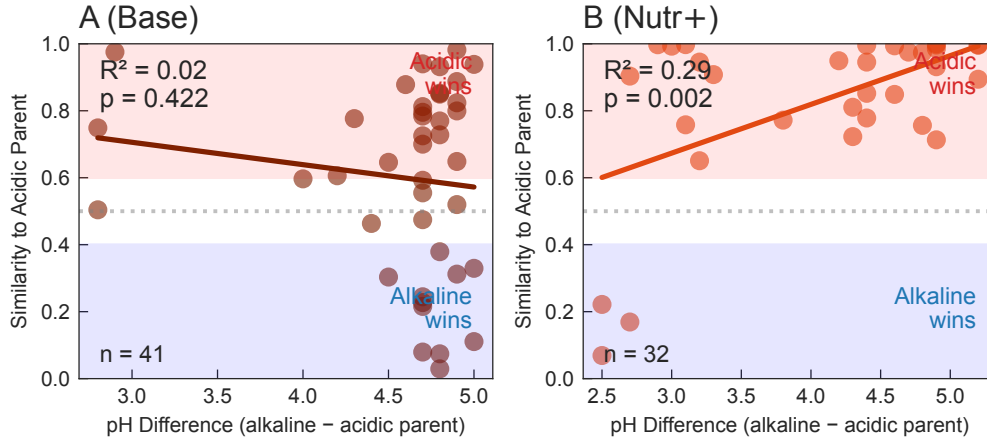
Extended Data Fig. 4. Effect of parental-community richness on coalescence outcomes. (a–e) Similarity maps showing coalescence outcomes for simulated parental communities with initial richness of 4, 6, 12, 24, and 48 species at $\mu = 0.6$. The qualitative pattern of Dominance-dominated outcomes is consistent across parental-community richness values. (f) Stacked bar plots showing outcome fractions (Dominance, Mixture, Restructuring) across parental-community richness values at three interaction-strength parameter values ($\mu = 0.3, 0.6, 0.8$). Dominance frequency remains relatively stable across parental-community richness values, while Mixture decreases and Restructuring increases with richer communities. Simulations used four non-overlapping parental communities per replicate, so total species-pool size scaled with parental-community richness ($N = 16\text{--}192$), with 200 independent replicates per condition and uniform interaction coefficients $\alpha_{ij} \sim U[0, 2\mu]$. See Supplementary Note 3 for the full sensitivity analysis.



Extended Data Fig. 5. Pairwise selection correlation increases with the interaction-strength parameter in simulations. (a–c) Pairwise selection correlation metric, computed from species-pair fate concordance, for species pairs from the same parental community (red, “Same parental community”) versus cross-community pairs (blue, “Cross-community”) at three representative interaction-strength parameter values. Gray horizontal lines indicate the random-selection baseline from a parental-affiliation-shuffle null model. Individual dots show per-event concordance values (100 stratified events displayed for legibility); squares with error bars show mean $\pm$ s.e.m. At low interaction-strength parameter value ($\mu = 0.3$), both within-community and cross-community pairs show positive concordance with a small difference ($\Delta = 0.05$). As μ increases ($\mu = 0.6, 0.8$), within-community pairs maintain positive concordance while cross-community pairs become increasingly negative ($\Delta = 0.59$ and 0.94), consistent with stronger parental-affiliation-correlated species fates. Two-sided permutation tests with 1,000 parental-affiliation label permutations gave $p < 0.001$ for Δ at all three displayed μ values. (d) Mean pairwise selection correlation across all simulated coalescence events as a function of the interaction-strength parameter μ . Species pairs from the same parental community (red circles) show positive concordance that increases with μ , while cross-community pairs (blue squares) show negative concordance that becomes more negative with μ . Error bars represent event-level s.e.m. across $n = 1,200$ coalescence events per interaction-strength parameter value generated from 200 independent replicates, with six pairwise coalescence events per replicate.

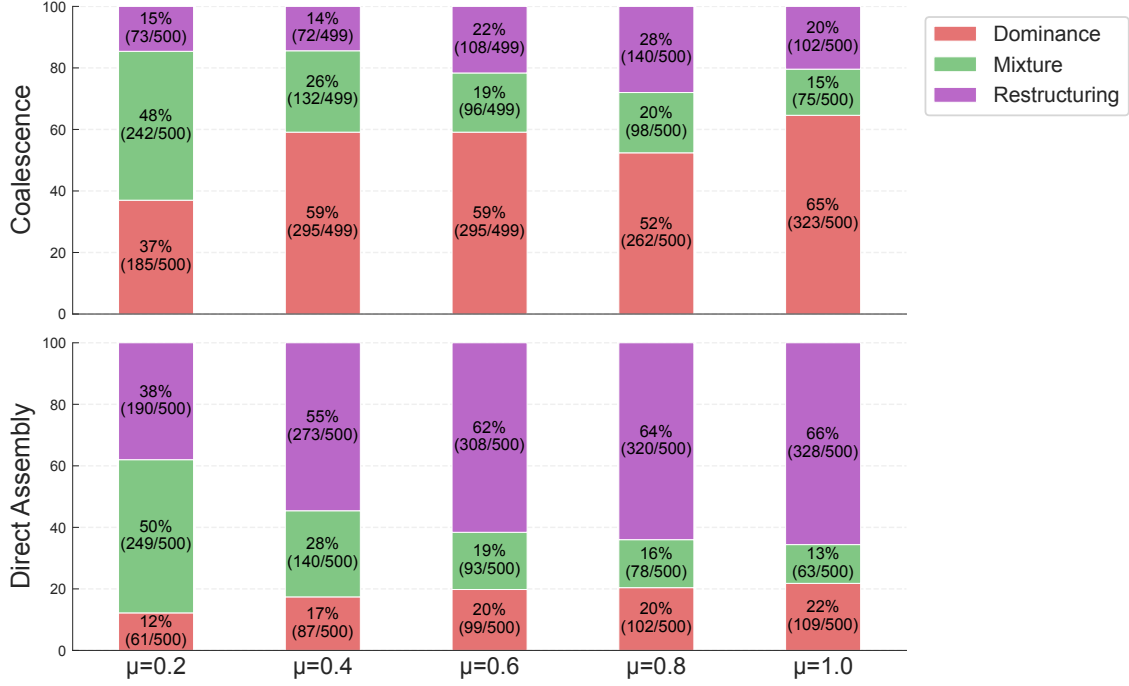


Extended Data Fig. 6. Pairwise selection correlation in experimental coalescence across nutrient conditions. Mean pairwise selection correlation metric, computed from species-pair fate concordance, for species pairs from the same parental community (red) versus cross-community pairs (blue). Gray horizontal line indicates the random-selection baseline from a parental-affiliation-shuffle null model. Individual dots show per-event concordance values; squares with error bars show mean $\pm$ s.e.m. **(a)** In Nutr $-$ medium, where failed-invasion frequency among the assayed abundant isolates is low, within-community and cross-community pairs show no significant difference in fate concordance ($n = 90$ valid events; $\Delta = 0.016$; two-sided permutation test with 1,000 parental-affiliation label permutations, $p = 0.307$), consistent with species-level dynamics and no detectable community-level selection by this metric. **(b)** In Base medium, the same-versus-cross separation is significant and largest among the three media ($n = 83$; $\Delta = 0.235$; $p < 0.001$). **(c)** In Nutr $+$ medium, where failed-invasion frequency and environmental feedbacks are higher, within-community species pairs exhibit significantly higher selection correlation than cross-community pairs ($n = 83$; $\Delta = 0.141$; $p < 0.001$), consistent with parental-affiliation-correlated species fates associated with community-level selection.



Extended Data Fig. 7. Parental-community pH difference is associated with Dominance direction in Nutr+. In the strict acidic-alkaline subset used for this PDI-regression analysis, where one parental community is acidic (pH < 6.5) and the other alkaline (pH > 7.5), pH difference is associated with the identity of the dominant parental community in Nutr+ but not in Base medium. **(a)** Base medium ($n = 41$; acidic community dominated in 23/41 events, 56.1%; $R^2 = 0.02$, $p = 0.42$, n.s.) and **(b)** Nutr+ medium ($n = 32$; acidic community dominated in 29/32 events, 90.6%; $R^2 = 0.29$, $p = 0.002$). X-axis: pH difference (alkaline parental community pH – acidic parental community pH). Y-axis: PDI transformed so larger values indicate greater contribution from the acidic parental community. Red shaded region indicates acidic community dominance (PDI > 0.6). Blue shaded region indicates alkaline community dominance (PDI < 0.4).

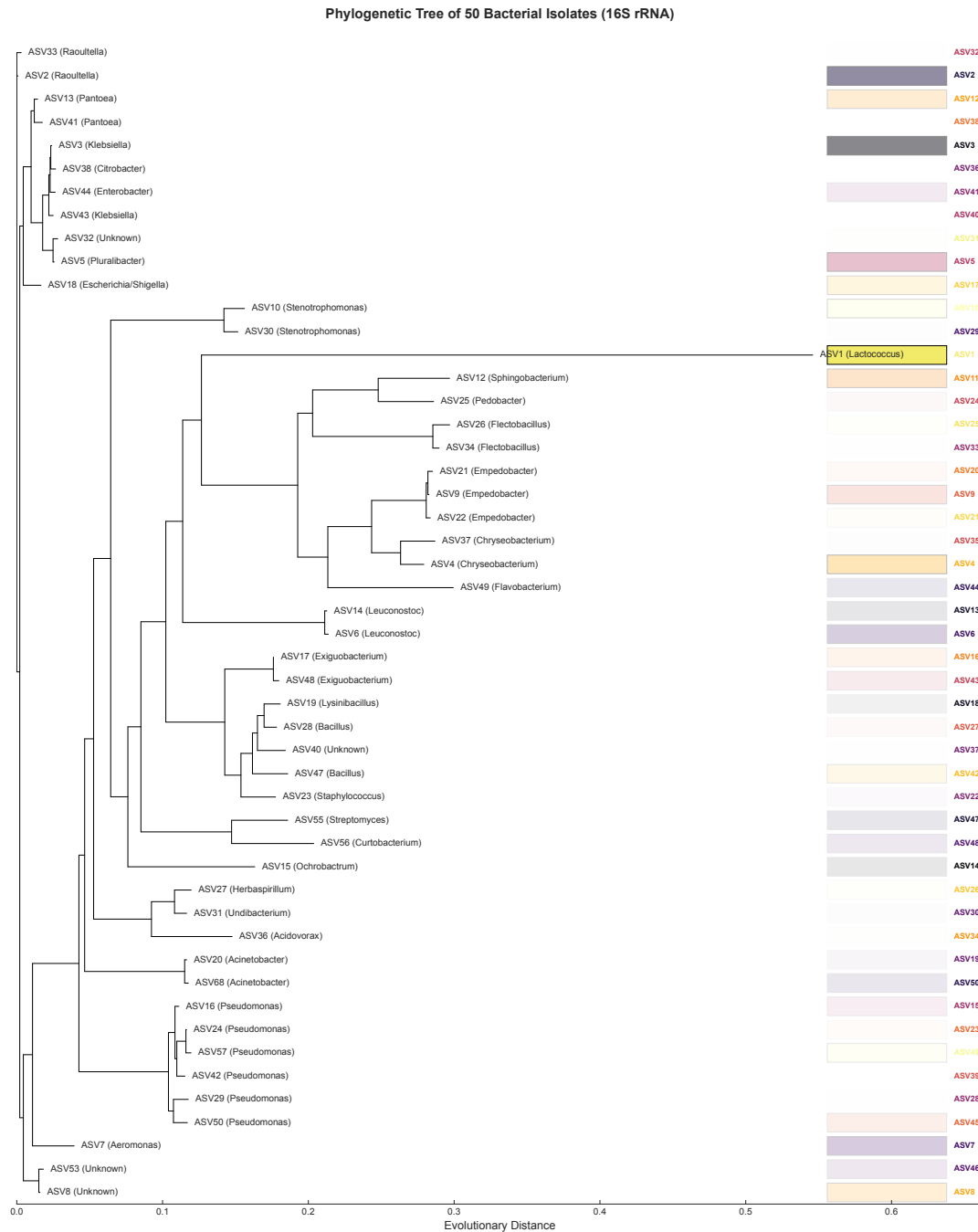
Assembly History Effect on Coalescence Outcomes (Simulation)



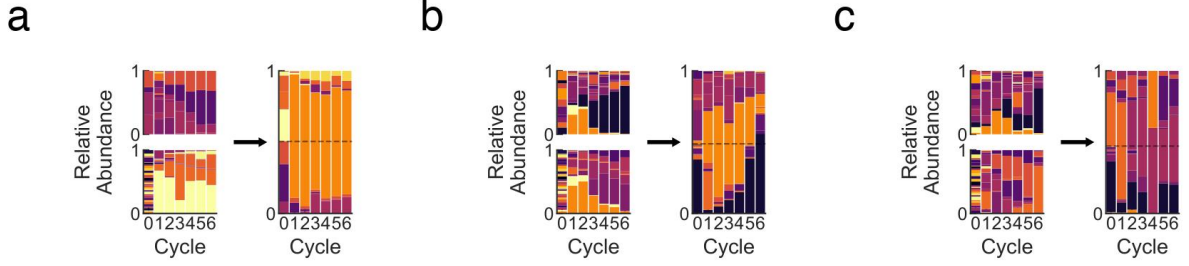
Extended Data Fig. 8. Assembly history promotes Dominance in simulations. This comparison tests whether assembly history itself, not the interaction-strength parameter alone, promotes Dominance by pre-filtering species before coalescence. Stacked bar plots comparing outcome distributions between coalescence of pre-assembled parental communities (top row) and direct assembly from the mixed species pool (bottom row) across five interaction-strength parameter values ($\mu = 0.2, 0.4, 0.6, 0.8, 1.0$). Percentages and counts (n/total) are shown for each category. Coalescence consistently produces higher Dominance and lower Restructuring compared to direct assembly, supporting the interpretation that assembly history contributes to parental-affiliation-correlated persistence.

Supplementary Figures

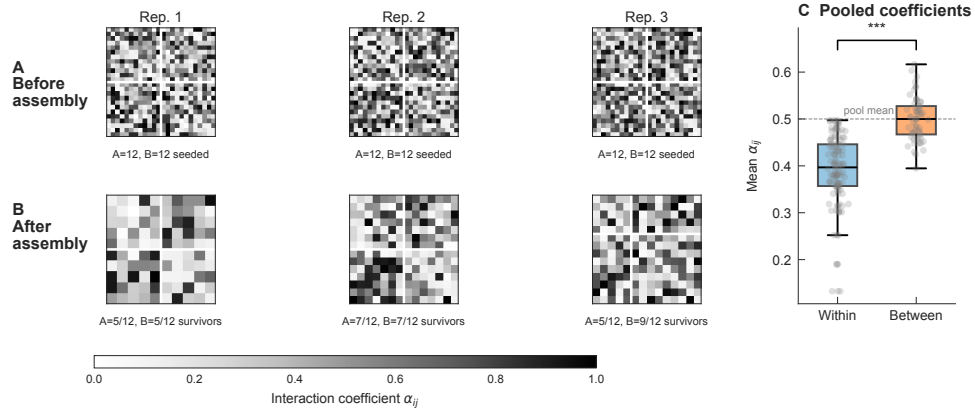
Strain library, coalescence examples, and assembly filtering



Supplementary Fig. 1. Phylogenetic tree of bacterial isolates. Maximum-likelihood phylogenetic tree based on full-length 16S rRNA gene sequences of the 54 bacterial isolates used in this study. Tree was constructed using EMBL-EBI tools³. The isolates span three phyla (Proteobacteria, Firmicutes, Bacteroidota) and 29 families, representing broad phylogenetic diversity.

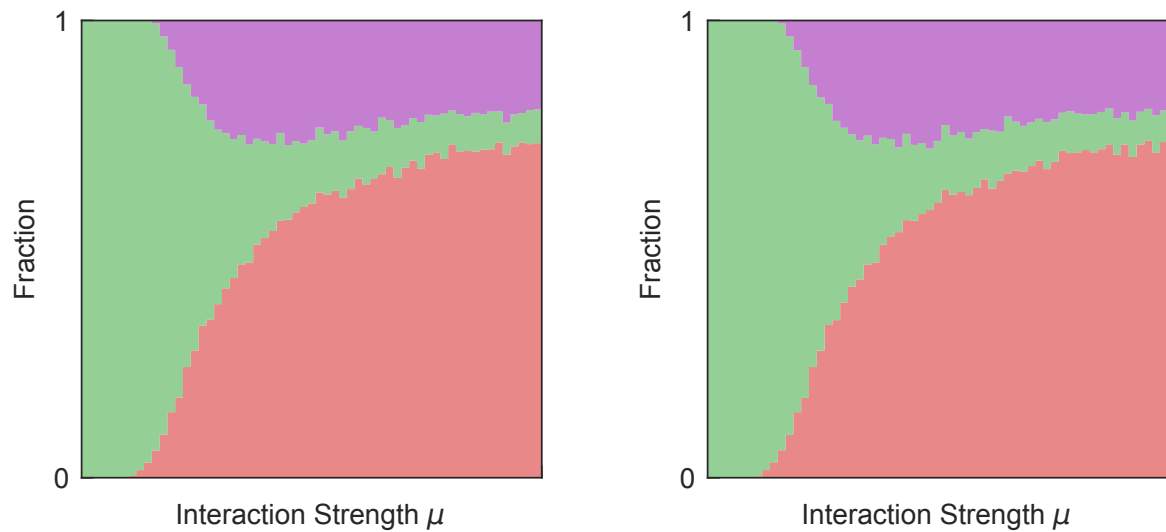


Supplementary Fig. 2. Time series of community composition during coalescence in Base medium. Each panel (a–c) shows two parental communities (left, stacked vertically) and their coalesced outcome (right) over serial dilution cycles. Stacked bar charts represent relative species abundances at each time point. X-axis: serial dilution cycles (0–7); Y-axis: relative abundance. These examples show Dominance and Restructuring dynamics, where one parental community rapidly displaces the other or the merged community converges toward a novel state.



Supplementary Fig. 3. Assembly filters simulated communities into groups of weak mutual competitors. **A**, Interaction-coefficient submatrices for two initially seeded parental communities before assembly in three example simulations at $\mu = 0.50$. Each parental community starts with 12 species. White lines separate the two parental-community species sets. **B**, Corresponding post-assembly submatrices containing only surviving species, ordered by parental-community origin. Labels below each matrix show the number of surviving species from each parent. **C**, Across all 10 full-matrix simulations at $\mu = 0.50$ (60 parental-community pairs), mean within-community coefficients among survivors were lower than between-community coefficients (0.389 vs 0.500; two-sided Mann–Whitney U test, $U = 624$, $p = 1.72 \times 10^{-19}$). Thus, assembly reduces mutual competition among co-assembled survivors while cross-parental coefficients remain near the pool mean.

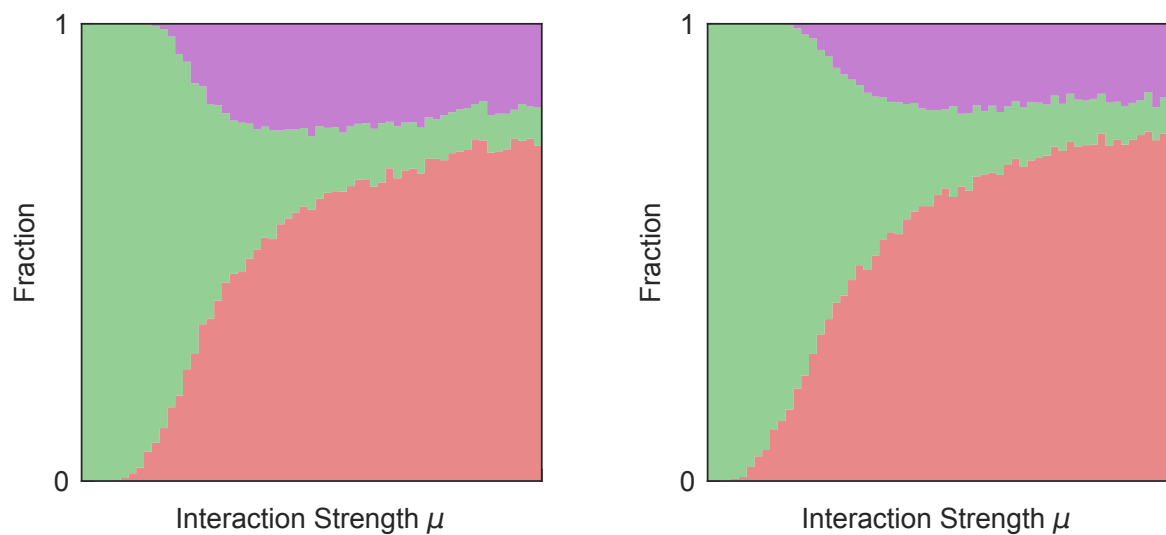
Baseline gLV parameter sensitivity



(a) Growth rate: mean 1, standard deviation 0.1

(b) Growth rate: mean 1, standard deviation 0.2

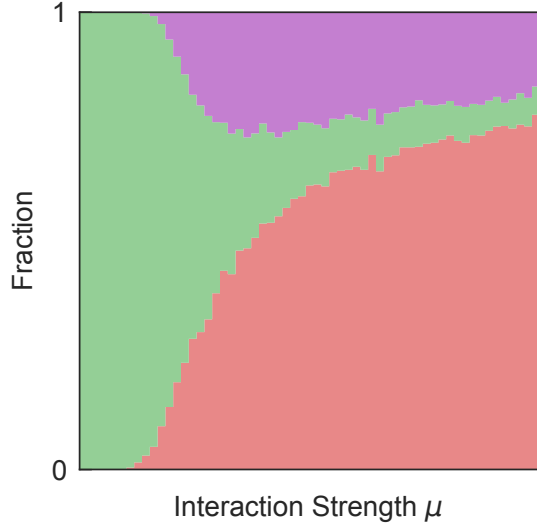
Supplementary Fig. 4. Phase diagrams with growth-rate heterogeneity. Coalescence outcomes when species have heterogeneous intrinsic growth rates sampled from normal distributions with mean 1 and standard deviations of (a) 0.1 and (b) 0.2. The qualitative transition from Mixture-dominated to Dominance/Restructuring outcomes with increasing interaction-strength parameter μ persisted under growth-rate variation.



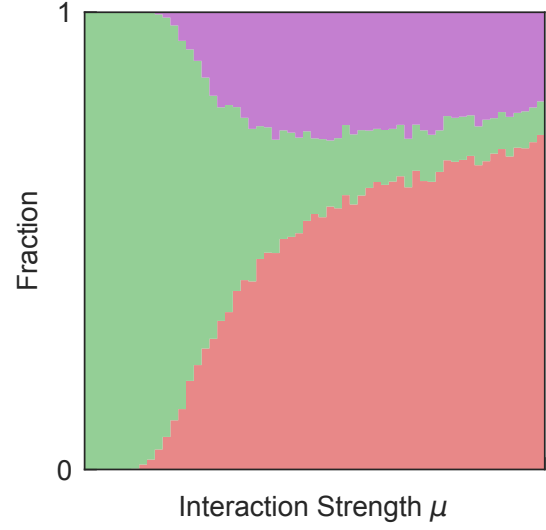
(a) Carrying capacity: mean 1, standard deviation 0.1

(b) Carrying capacity: mean 1, standard deviation 0.2

Supplementary Fig. 5. Phase diagrams with carrying-capacity variation. Coalescence outcomes when species have heterogeneous carrying capacities sampled from normal distributions with mean 1 and standard deviations of (a) 0.1 and (b) 0.2. The qualitative transition from Mixture-dominated to Dominance/Restructuring outcomes with increasing interaction-strength parameter μ persisted under carrying-capacity variation.



(a) Gaussian distribution (mean μ , matched variance)



(b) Gamma distribution (mean μ , matched variance)

Supplementary Fig. 6. Phase diagrams for alternative interaction coefficient distributions. The Gaussian and Gamma distributions are both matched to the uniform baseline mean μ and variance $\mu^2/3$. **(a)** Gaussian distribution, which can include negative coefficients. **(b)** Gamma distribution, which is inherently non-negative. The qualitative transition from Mixture-dominated to Dominance/Restructuring outcomes with increasing interaction-strength parameter μ persisted under both alternative distributions, indicating that the transition is not specific to one coefficient distribution.

Nutrient-dependent invasion resistance assays



Supplementary Fig. 7. Pairwise invasion outcomes in Nutr– medium. Matrix showing competitive outcomes among the 12 most abundant isolates. Each cell indicates whether the invader (initially 5%) successfully established or was excluded after seven growth-dilution cycles. Low failed-invasion frequency among the assayed abundant isolates indicates relatively weak invasion resistance in this assay.

Pairwise Species 95:5 Invasion Assays — Base



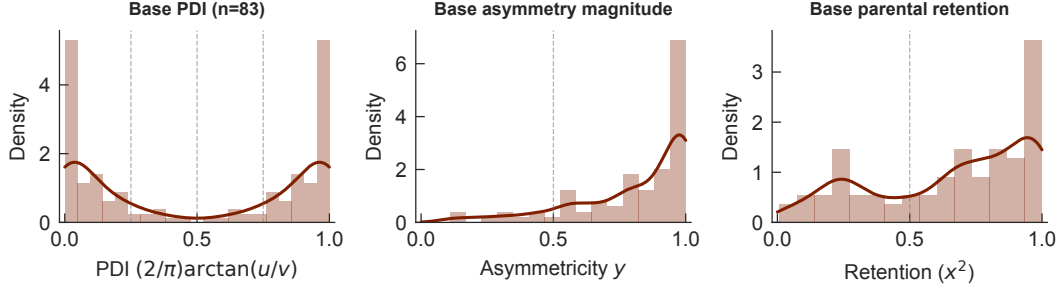
Supplementary Fig. 8. Pairwise invasion outcomes in Base medium. Matrix showing competitive outcomes among the 12 most abundant isolates. Each cell indicates whether the invader (initially 5%) successfully established or was excluded after seven growth-dilution cycles. Intermediate failed-invasion frequency among the assayed abundant isolates indicates intermediate invasion resistance in this assay.

Pairwise Species 95:5 Invasion Assays — Nutr+

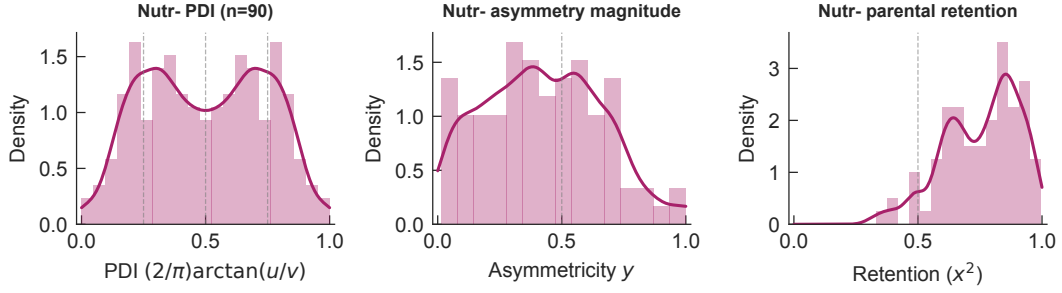


Supplementary Fig. 9. Pairwise invasion outcomes in Nutr+ medium. Matrix showing competitive outcomes among the 12 most abundant isolates. Each cell indicates whether the invader (initially 5%) successfully established or was excluded after seven growth-dilution cycles. High failed-invasion frequency among the assayed abundant isolates indicates stronger invasion resistance in this assay.

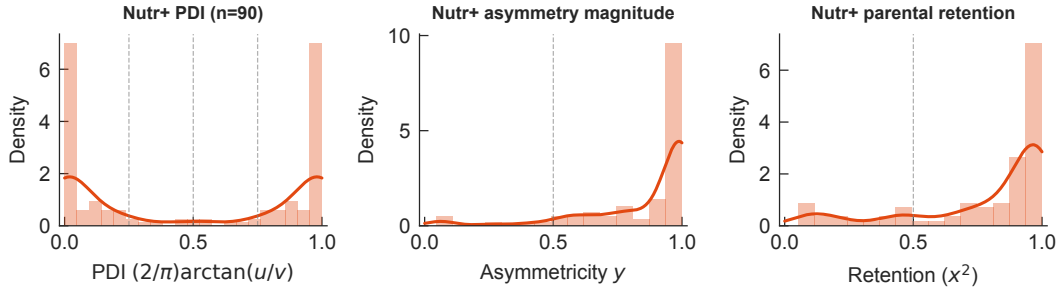
Continuous outcome-classification metrics



Supplementary Fig. 10. Base-medium continuous similarity measures underlying categorical outcome classification. From left to right: PDI distribution, direction-independent parental asymmetry $|2\text{PDI} - 1|$, and squared retention magnitude $r^2 = u^2 + v^2$ for the Base-medium coalescence outcomes shown in Fig. 1E. These continuous distributions make the categorical Base-medium classifications transparent by showing the underlying asymmetry and retention values. PDI is computed as $(2/\pi)\arctan(u/v)$, with PDI defined as 1 for the boundary case $v = 0$ and $u > 0$.

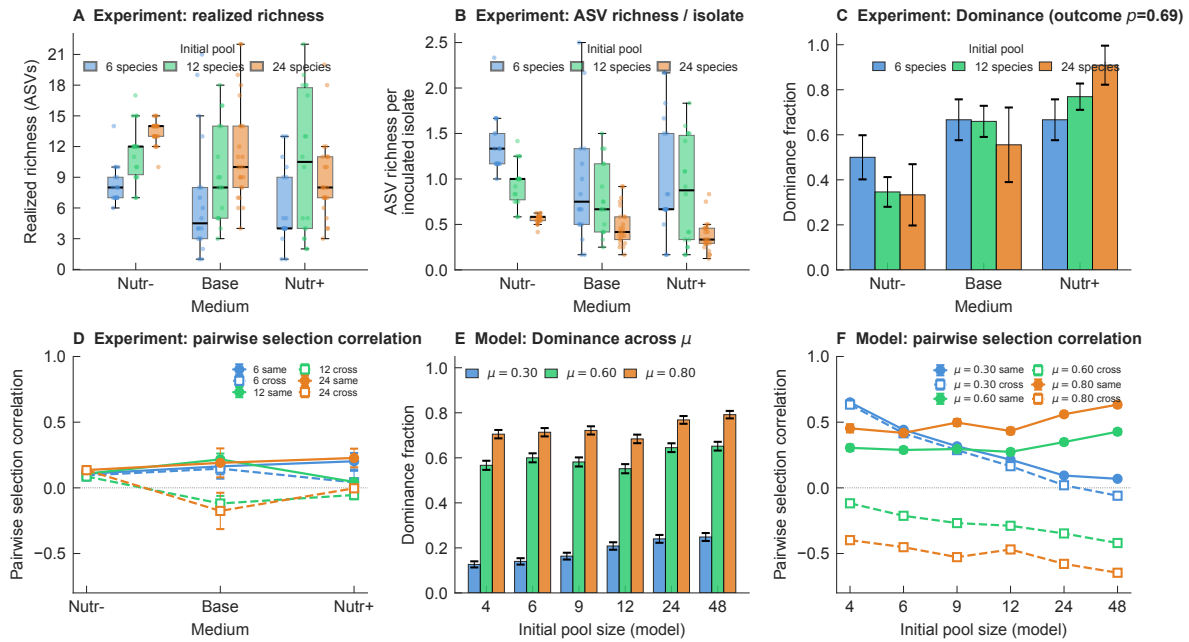


Supplementary Fig. 11. Nutr- continuous similarity measures underlying categorical outcome classification. From left to right: PDI distribution, direction-independent parental asymmetry $|2\text{PDI} - 1|$, and squared retention magnitude $r^2 = u^2 + v^2$. PDI is computed as $(2/\pi)\arctan(u/v)$, with PDI defined as 1 for the boundary case $v = 0$ and $u > 0$.

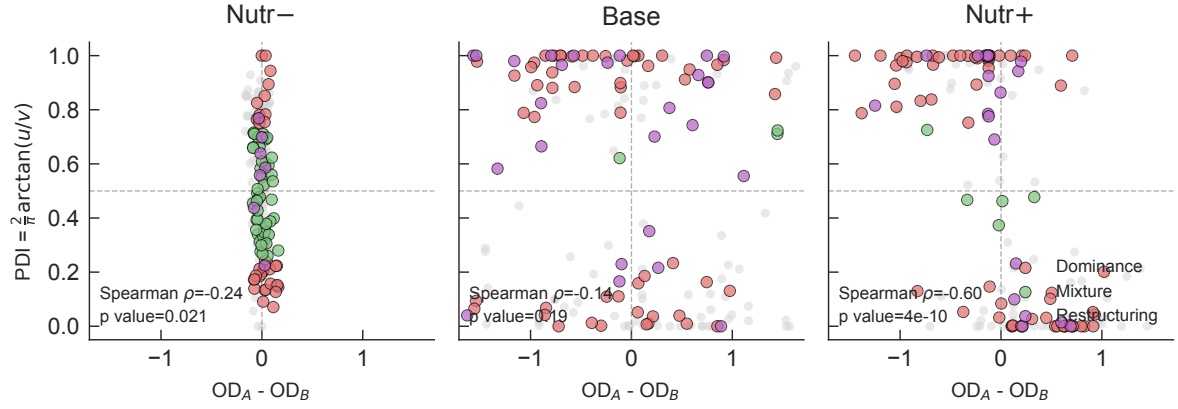


Supplementary Fig. 12. Nutr+ continuous similarity measures underlying categorical outcome classification. From left to right: PDI distribution, direction-independent parental asymmetry $|2\text{PDI} - 1|$, and squared retention magnitude $r^2 = u^2 + v^2$. PDI is computed as $(2/\pi)\arctan(u/v)$, with PDI defined as 1 for the boundary case $v = 0$ and $u > 0$.

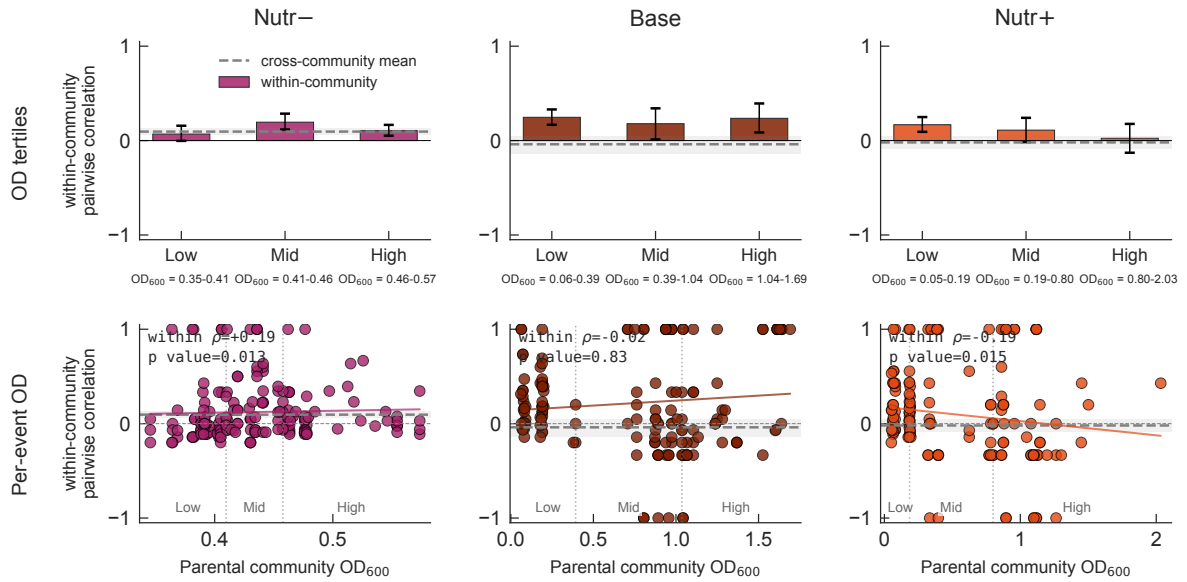
Pool-size, biomass, and dominant-species controls



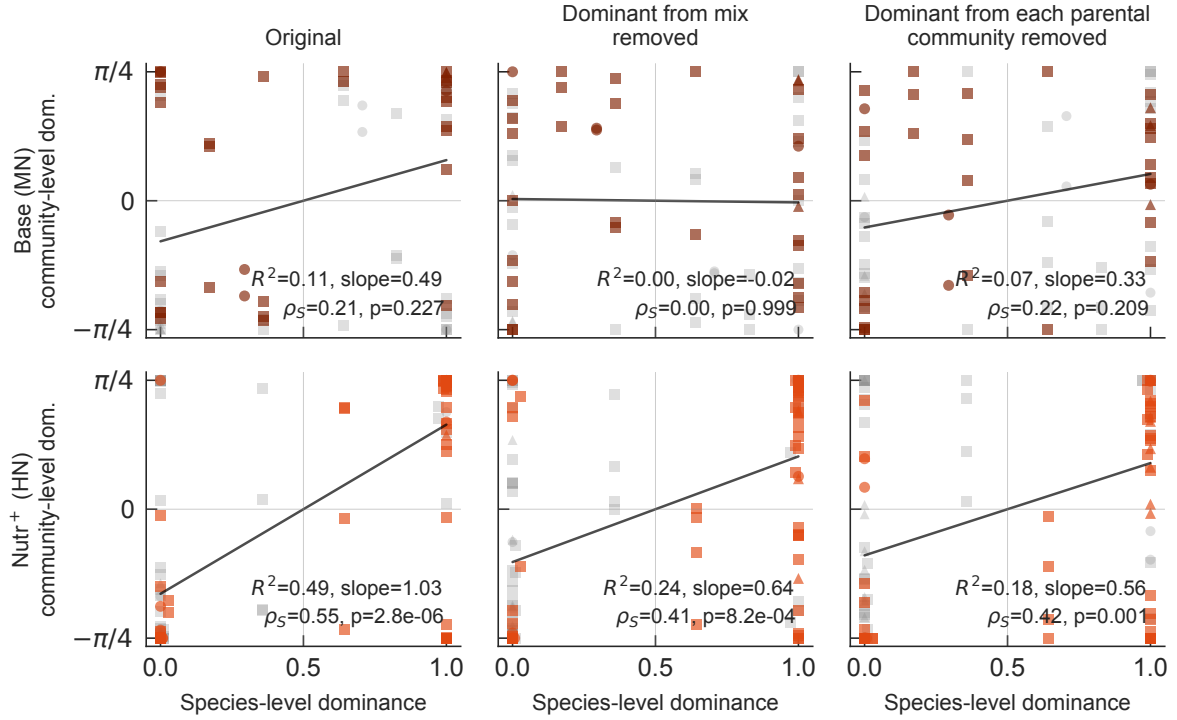
Supplementary Fig. 13. Dominance frequency is primarily associated with nutrient condition or model interaction-strength parameter, rather than initial pool size. (a) Realized parental richness (ASVs above 0.1% threshold) shown by medium, with boxplots grouped and colored by initial inoculated isolate richness; the pooled statistic uses a Kruskal–Wallis test of richness across pool sizes. (b) ASV richness per inoculated isolate, used as an assembly-retention proxy and defined as realized parental ASV richness divided by inoculated isolate richness, shown by medium and initial inoculated isolate richness; the pooled statistic uses a Kruskal–Wallis test across pool sizes. (c) Experimental Dominance fraction shown by medium, with bars grouped and colored by initial inoculated isolate richness; the displayed pooled p -value is from a chi-square test of independence for the full three-category outcome distribution across pool sizes, not a binary Dominance-only test. (d) Experimental pairwise selection correlation metric grouped by initial inoculated isolate richness; solid circles indicate same-parental-community species pairs and dashed open squares indicate cross-parental-community species pairs, using the same fate-concordance implementation as Fig. 2D. (e) Model Dominance fraction grouped by μ (600 coalescence pairs per bar). (f) Model pairwise selection correlation metric (same solid / cross dashed) at $\mu \in \{0.30, 0.60, 0.80\}$ across parental-community richness values 4–48; total simulated species-pool size scales as four communities per replicate ($N = 16$ –192). Panels (d)–(f) summarize observed trends descriptively; no additional p -values are shown for these panels.



Supplementary Fig. 14. Continuous parental-community OD_{600} differences do not support the prediction that higher OD predicts parental dominance direction. Signed ΔOD_{A-B} ($OD_A - OD_B$) vs PDI is shown per medium. Spearman ρ is the rank correlation between signed ΔOD_{A-B} and PDI; the displayed p value is from a two-sided Spearman rank-correlation test. Gray points are reflections included for symmetry.

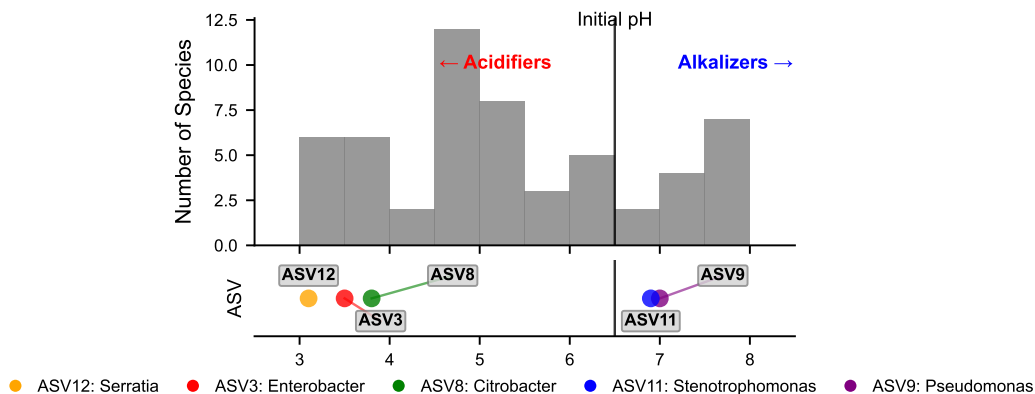


Supplementary Fig. 15. Higher parental-community OD_{600} does not produce a consistent increase in within-community pairwise selection correlation. Columns show Nutr-, Base, and Nutr+, respectively. Top row: parental communities grouped into low, mid, and high within-medium OD tertiles, with OD_{600} ranges shown below the group labels. Bars show within-community pairwise selection correlation with bootstrap 95% CIs; the dashed gray line and shaded band show the medium-level mean cross-community selection correlation and its bootstrap 95% CI. Bottom row: per-event parental-community observations plotted by OD and within-community pairwise selection correlation.

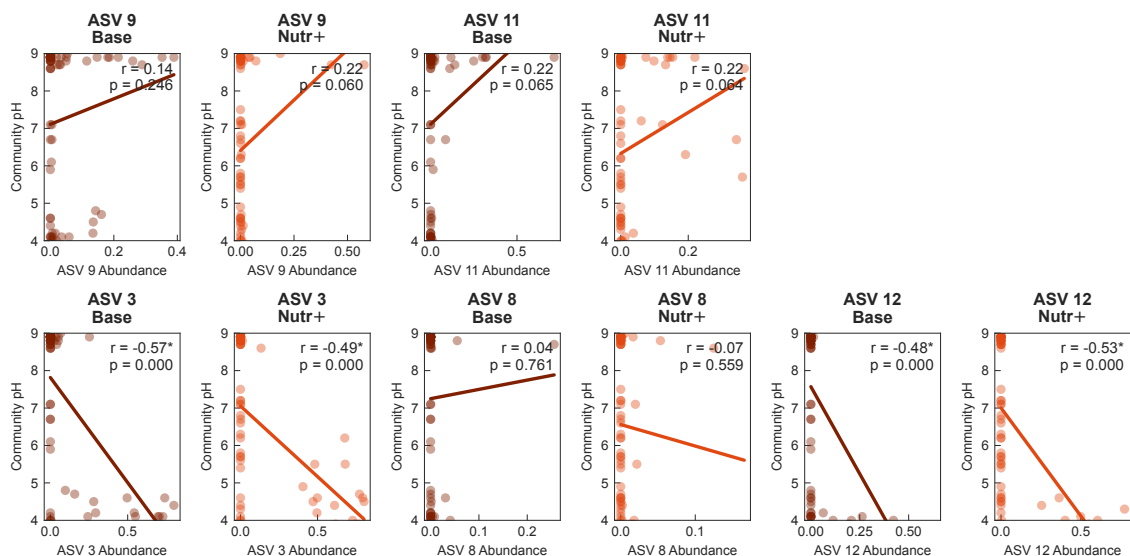


Supplementary Fig. 16. Dominant-species removal control for the Fig. 5C PDI analysis. Per-medium scatter plots are shown without pooling Base and Nutr+ events. Rows show Base and Nutr+; columns show the original Fig. 5C calculation, removal of the most abundant species in the coalesced community from all three community composition vectors, and removal of the most abundant species from each parental community from all three composition vectors before recalculating PDI. Independent-event sample sizes are Base $n = 34/34/34$ and Nutr+ $n = 63/62/55$ for original / coalesced-community dominant removed / parental-community dominants removed, respectively. The third column is the stricter test. R^2 and slope annotations are computed on the reflected plotting data, as in Fig. 5C, and are used as descriptive summaries. Spearman ρ and p are computed on the independent, non-reflected events.

Isolate pH phenotypes and parental-community pH structure



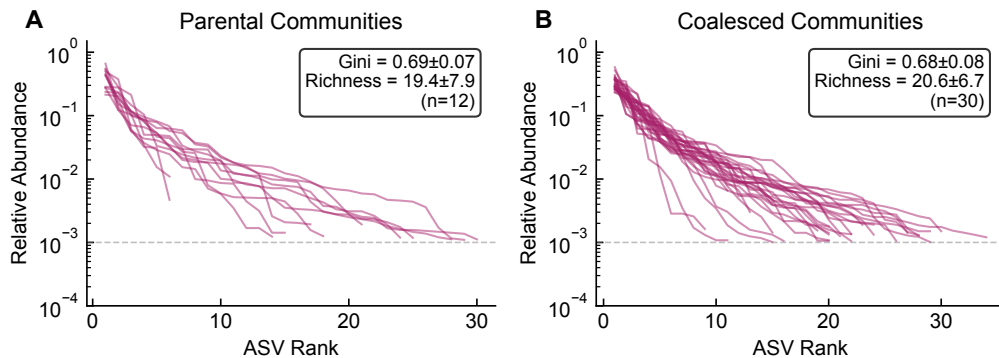
Supplementary Fig. 17. Monoculture pH modification by bacterial isolates. Distribution of monoculture pH after 15 h growth for all ASVs, showing the range of pH modification abilities relative to the initial pH of 6.5. Strong acidifiers (monoculture pH < 5) include ASV 12 (*Serratia*), ASV 3 (*Enterobacter*), and ASV 8 (*Citrobacter*). Species that maintain or slightly increase pH above the initial pH (monoculture pH > 6.5) include ASV 11 (*Stenotrophomonas*) and ASV 9 (*Pseudomonas*). This isolate-level pH phenotype provides the basis for later pH-feedback and pH-contrast analyses. Bottom legend shows taxa identities for the labeled ASVs.



Supplementary Fig. 18. Dominant species class (acidifier vs. alkalizer) is associated with parental community pH. Scatter plots showing the relationship between dominant ASV relative abundance in parental communities and the resulting community pH at stabilization. Top row: alkalizers including ASV 9 (*Pseudomonas*) and ASV 11 (*Stenotrophomonas*), whose dominance is associated with higher community pH. Bottom row: acidifiers including ASV 3 (*Enterobacter*), ASV 8 (*Citrobacter*), and ASV 12 (*Serratia*), whose dominance is associated with lower community pH. For each ASV, left panel shows Base medium and right panel shows Nutr+ medium. Regression lines and correlation coefficients are shown; asterisks indicate $p < 0.05$.

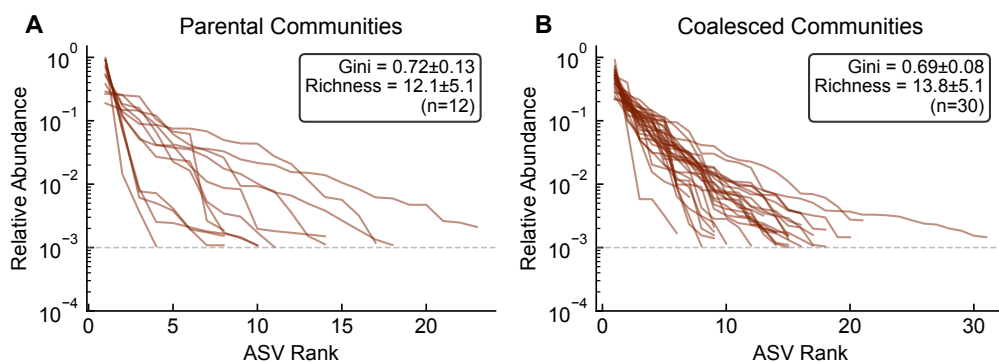
Laboratory-stabilized natural sample-derived communities

Nutr- - Natural Communities



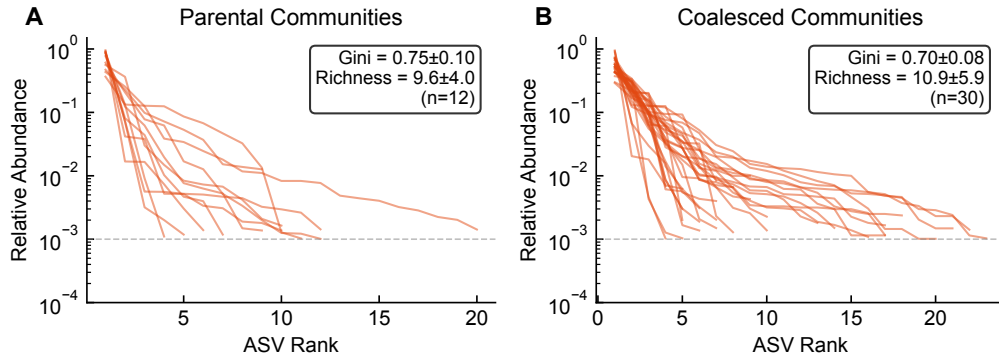
Supplementary Fig. 19. Rank-abundance curves for natural sample-derived communities in Nutr– medium. **(a)** Parental communities before coalescence. **(b)** Coalesced communities after coalescence. Each line represents one community’s rank-abundance curve. Gini coefficients quantify abundance inequality (0 = perfectly even, 1 = one species dominates). ASV richness indicates number of ASVs above 0.1% relative abundance threshold. Horizontal dashed line indicates 0.1% threshold. Laboratory-stabilized communities derived from environmental samples retained higher ASV richness compared to synthetic communities (which had controlled initial inoculated richness of 6, 12, or 24 isolates). These rank-abundance and richness summaries provide context for comparing natural-community structure with the controlled synthetic-community design.

Base - Natural Communities

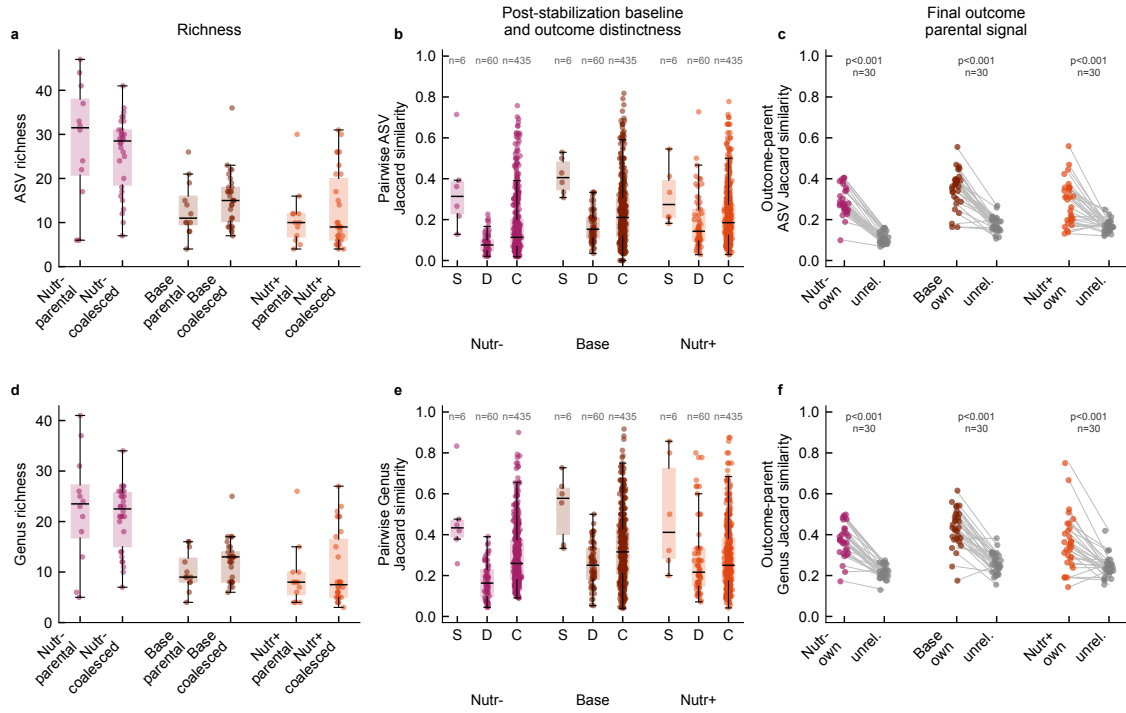


Supplementary Fig. 20. Rank-abundance curves for natural sample-derived communities in Base medium (natural-community analog of Supplementary Fig. 24 for synthetic communities). **(a)** Parental communities before coalescence. **(b)** Coalesced communities after coalescence. Each line represents one community’s rank-abundance curve. Gini coefficients quantify abundance inequality (0 = perfectly even, 1 = one species dominates). ASV richness indicates number of ASVs above 0.1% relative abundance threshold. Horizontal dashed line indicates 0.1% threshold.

Nutr+ - Natural Communities



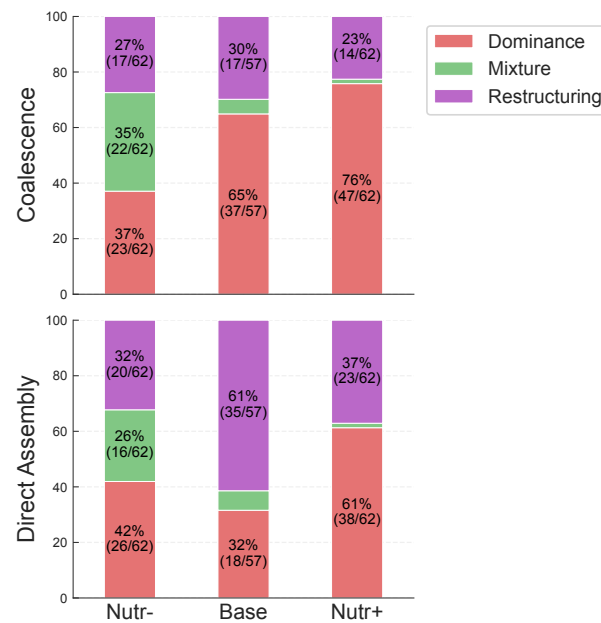
Supplementary Fig. 21. Rank-abundance curves for natural sample-derived communities in Nutr+ medium (natural-community analog of Supplementary Fig. 26 for synthetic communities). **(a)** Parental communities before coalescence. **(b)** Coalesced communities after coalescence. Each line represents one community's rank-abundance curve. Gini coefficients quantify abundance inequality (0 = perfectly even, 1 = one species dominates). ASV richness indicates number of ASVs above 0.1% relative abundance threshold. Horizontal dashed line indicates 0.1% threshold.



Supplementary Fig. 22. Natural sample-derived communities retain ASV- and genus-level distinctness after stabilization and coalescence. **a,d**, ASV- and genus-level richness by medium. **b,e**, Stabilized parental replicates from the same environmental source sample are more similar than parental communities from different source samples, indicating retained source-specific taxonomic signal after stabilization. **c,f**, Coalesced communities remain more similar to their own parental communities than to unrelated same-medium parental communities. Analyses use ASVs above the 0.1% relative-abundance threshold and post-stabilization 16S profiles; they test for complete post-stabilization taxonomic convergence to a single shared endpoint and retained parent-specific signal, but not pre-to-post enrichment convergence or functional convergence.

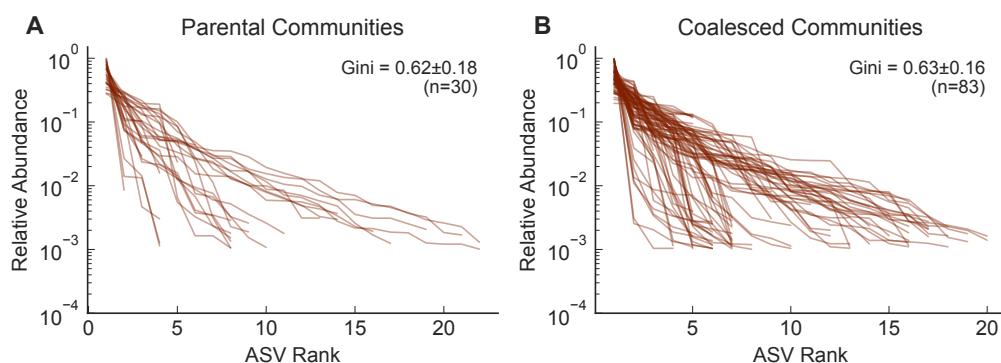
Assembly-history and abundance-structure controls

Assembly History Effect on Coalescence Outcomes (Experimental)



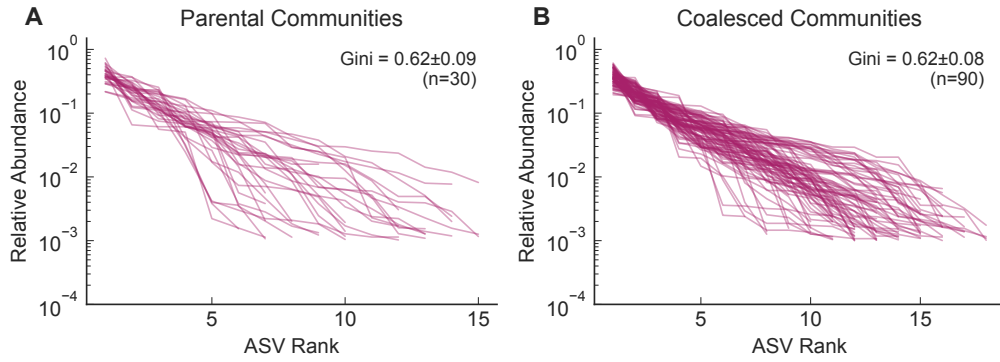
Supplementary Fig. 23. Assembly history promotes Dominance in experiments. Stacked bar plots comparing outcome distributions between coalescence (top row) and direct assembly (bottom row) across three nutrient conditions (Nutr-, Base, Nutr+). Percentages and counts (n/total) are shown for each category. In Base and Nutr+ conditions, coalescence shows elevated Dominance rates compared to direct assembly, consistent with simulation predictions (Extended Data Fig. 8, Supplementary Fig. 32).

Base



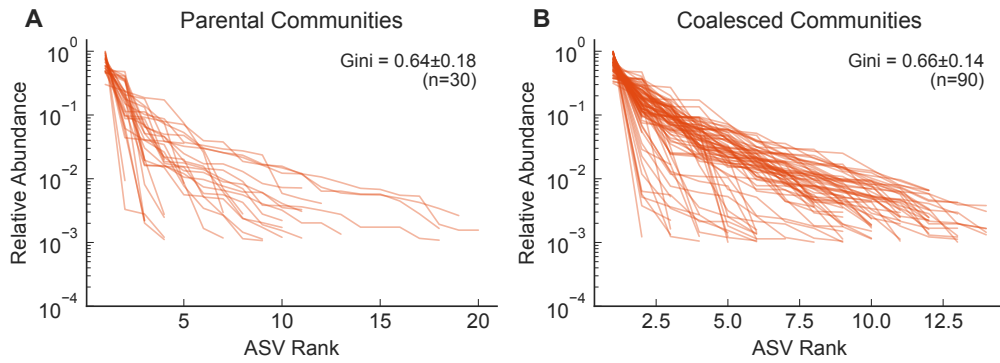
Supplementary Fig. 24. Rank-abundance curves comparing parental and coalesced communities in Base medium. (a) Parental communities before coalescence. (b) Coalesced communities after coalescence. Each line represents one community's rank-abundance curve. Gini coefficients quantify abundance inequality (0 = perfectly even, 1 = one species dominates). Parental: Gini = 0.62 ± 0.18 (n=30); Coalesced: Gini = 0.63 ± 0.16 (n=83). Both parental and coalesced communities exhibit substantial abundance skewness with similar Gini values.

Nutr-



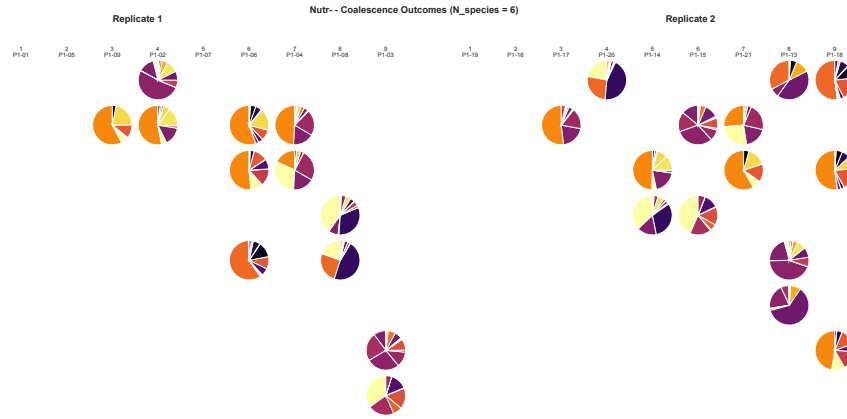
Supplementary Fig. 25. Rank-abundance curves comparing parental and coalesced communities in Nutr– medium. **(a)** Parental communities before coalescence. **(b)** Coalesced communities after coalescence. Each line represents one community’s rank-abundance curve. Parental: Gini = 0.62 ± 0.09 (n=30); Coalesced: Gini = 0.62 ± 0.08 (n=90).

Nutr+

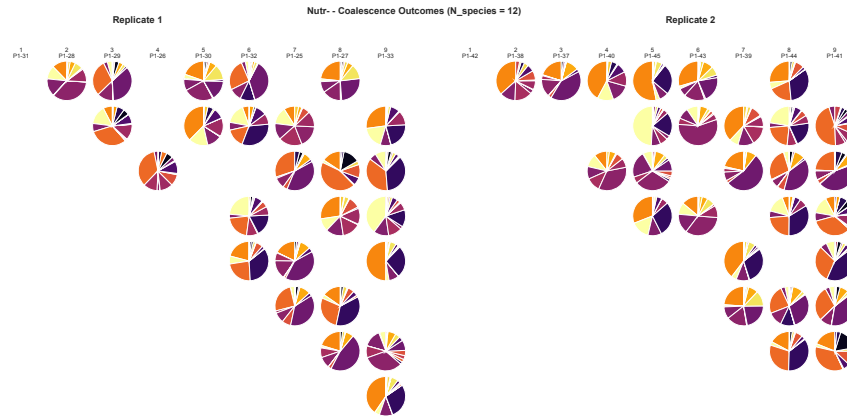


Supplementary Fig. 26. Rank-abundance curves comparing parental and coalesced communities in Nutr+ medium. **(a)** Parental communities before coalescence. **(b)** Coalesced communities after coalescence. Each line represents one community’s rank-abundance curve. Parental: Gini = 0.64 ± 0.18 (n=30); Coalesced: Gini = 0.66 ± 0.14 (n=90).

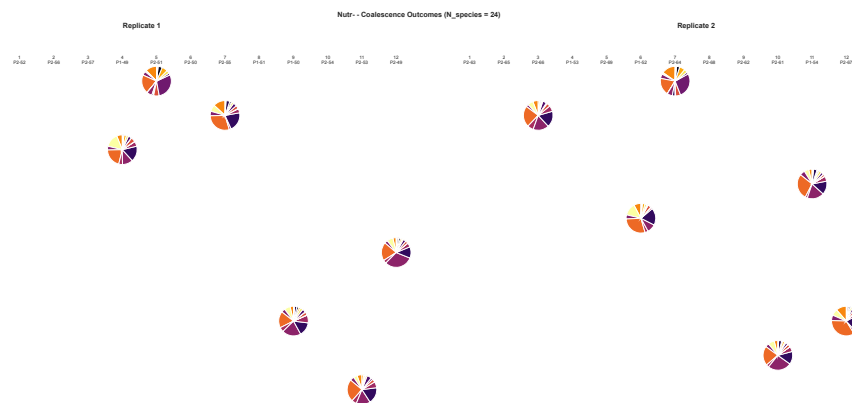
Nutrient-condition coalescence matrices and time series



(a) Initial richness: 6 species

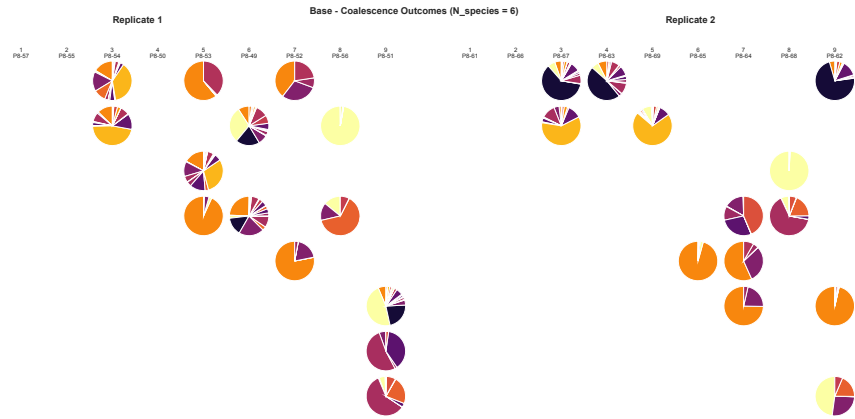


(b) Initial richness: 12 species

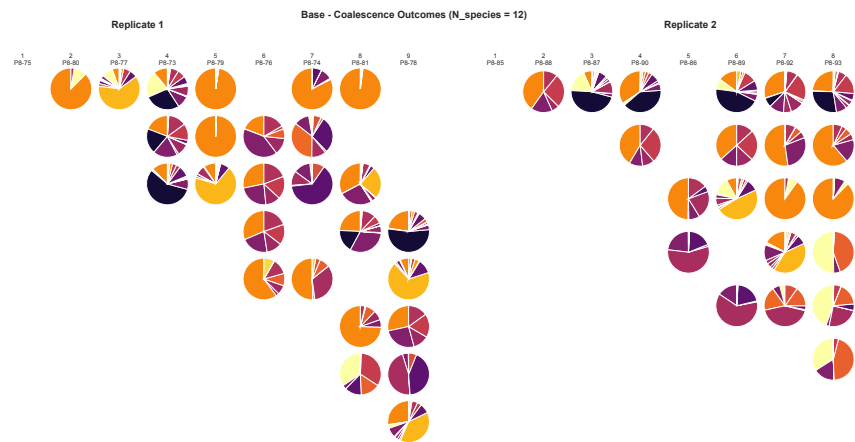


(c) Initial richness: 24 species

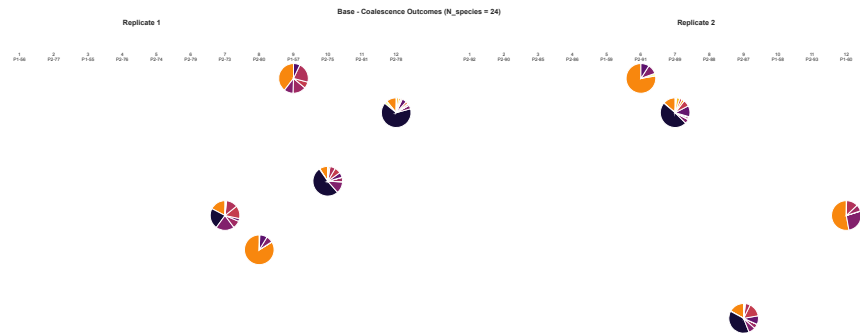
Supplementary Fig. 27. Coalescence outcome matrices in Nutr- medium. Each matrix shows pairwise coalescence outcomes for all parental community combinations at different initial inoculated richness levels. Pie charts display the species composition of coalesced communities; colors match parental origins. In this medium, which has lower failed-invasion frequency among the assayed abundant isolates, outcomes show more balanced mixing.



(a) Initial richness: 6 species

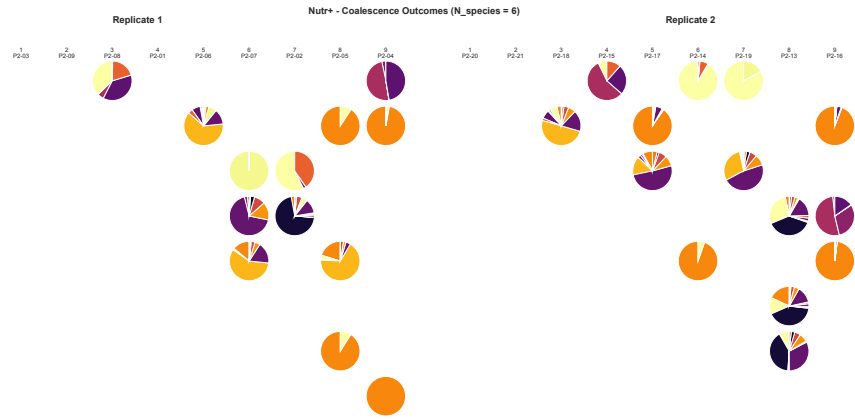


(b) Initial richness: 12 species

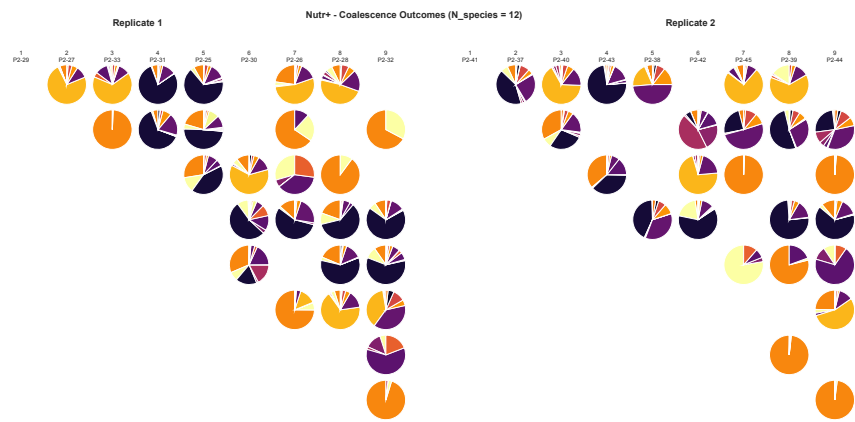


(c) Initial richness: 24 species

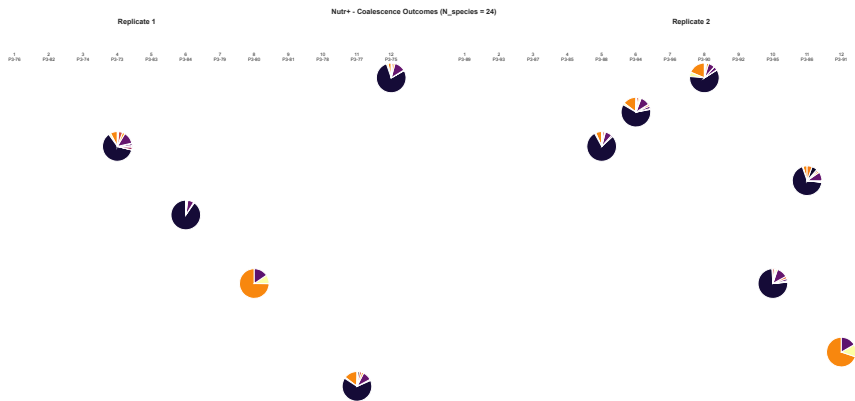
Supplementary Fig. 28. Coalescence outcome matrices in Base medium. Each matrix shows pairwise coalescence outcomes for all parental community combinations at different initial inoculated richness levels. Pie charts display the species composition of coalesced communities; colors match parental origins. In this medium, which has intermediate failed-invasion frequency among the assayed abundant isolates, one-sided outcomes (Dominance) occur frequently.



(a) Initial richness: 6 species

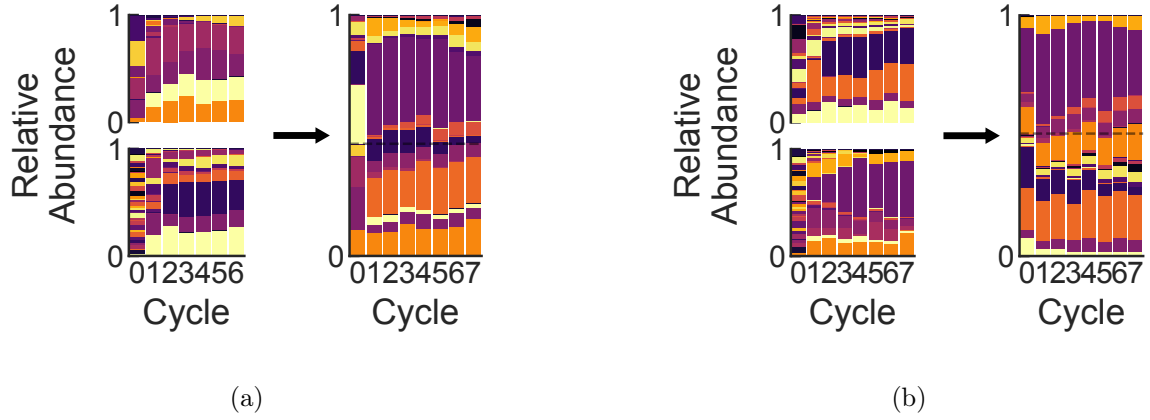


(b) Initial richness: 12 species

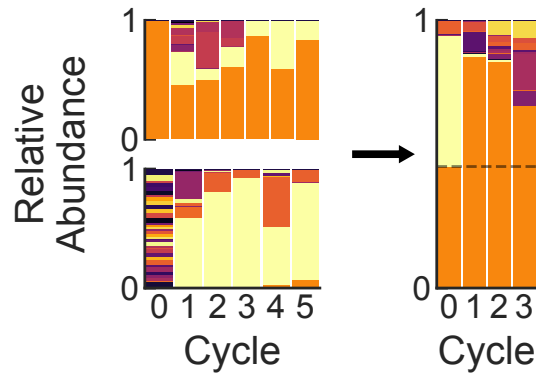


(c) Initial richness: 24 species

Supplementary Fig. 29. Coalescence outcome matrices in Nutr+ medium. Each matrix shows pairwise coalescence outcomes for all parental community combinations at different initial inoculated richness levels. Pie charts display the species composition of coalesced communities; colors match parental origins. In this medium, which has higher failed-invasion frequency among the assayed abundant isolates, one-sided outcomes (Dominance) predominate.

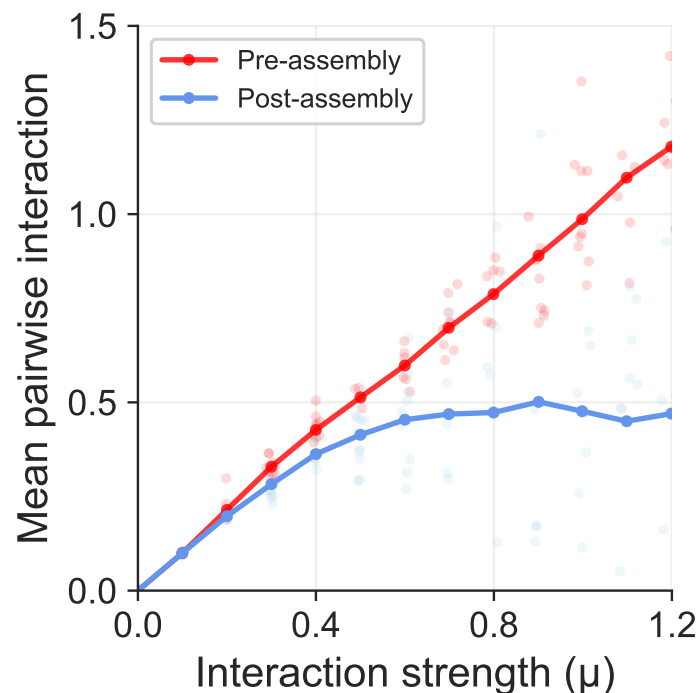


Supplementary Fig. 30. Time series of community composition during coalescence in Nutr– medium. Each panel (a, b) shows two parental communities (left, stacked vertically) and their coalesced outcome (right) over serial dilution cycles. Stacked bar charts represent relative species abundances at each time point. X-axis: serial dilution cycles (0–7); Y-axis: relative abundance.

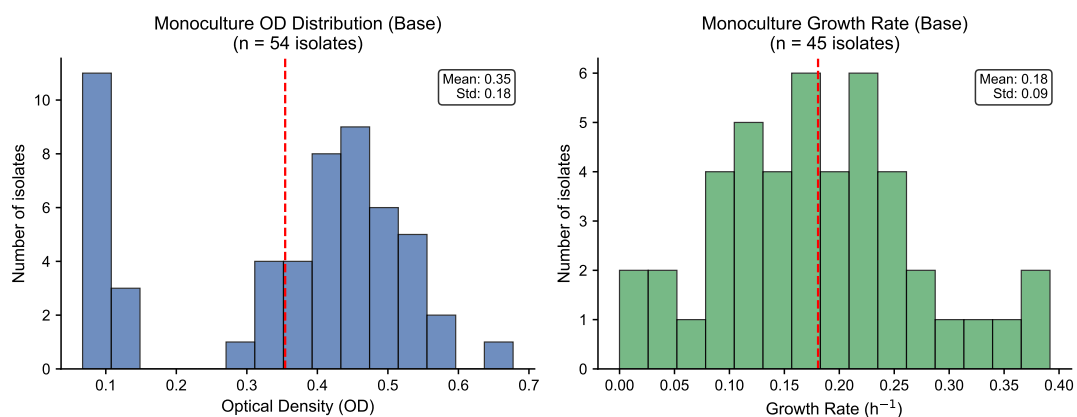


Supplementary Fig. 31. Time series of community composition during coalescence in Nutr+ medium. The panel shows two parental communities (left, stacked vertically) and their coalesced outcome (right) over serial dilution cycles. Stacked bar charts represent relative species abundances at each time point. X-axis: serial dilution cycles (0–7); Y-axis: relative abundance.

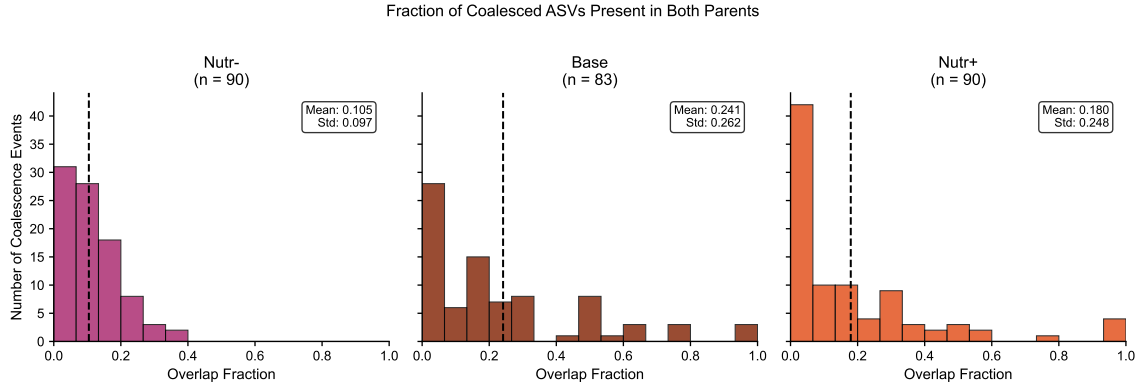
Assembly filtering, isolate phenotypes, and parental-overlap controls



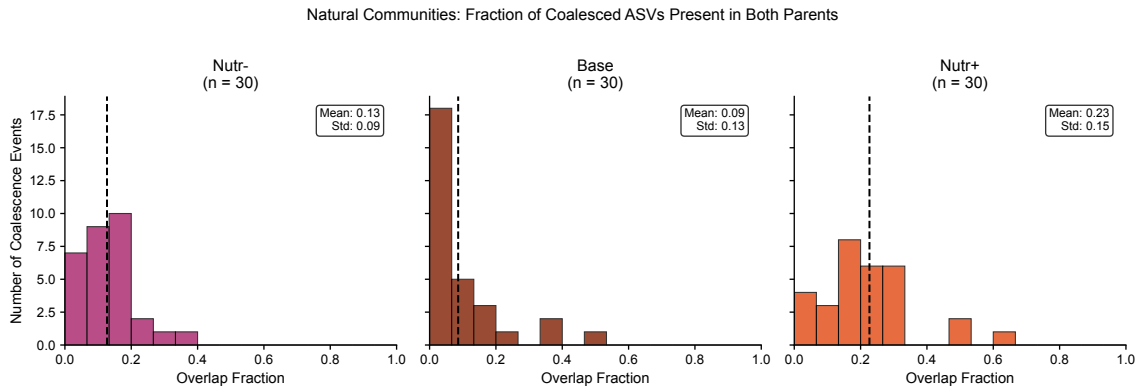
Supplementary Fig. 32. Assembly reduces average pairwise interaction coefficients in simulations. Average pairwise interaction coefficient before assembly (red, pre-assembly species pool) versus after assembly (blue, within-community) across different interaction-strength parameter values μ . Points show individual simulation runs; lines connect means. Post-assembly communities consistently show reduced average coefficients, indicating that assembly filters out strongly competing species.



Supplementary Fig. 33. Phenotypic diversity of bacterial isolates in monoculture (Base medium). **(a)** Distribution of optical density (OD_{600}) reached by bacterial isolates during monoculture growth in Base medium ($n = 54$ isolates). **(b)** Distribution of exponential growth rates across isolates ($n = 45$ isolates with measurable growth rates out of 54 total). Growth rates were computed by log-linear regression of OD versus time during the exponential phase from full kinetic time-course measurements (475 cycles, ~ 3 min intervals). The isolates exhibit broad variation in both growth yield and growth rate, reflecting phenotypic diversity that may contribute to varied coalescence outcomes in community mixing experiments.

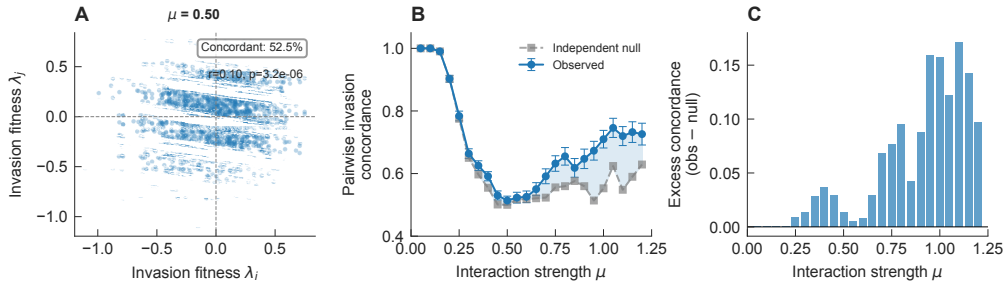


Supplementary Fig. 34. Fraction of coalesced ASVs present in both parental communities. Distribution of overlap fraction across coalescence events in three nutrient conditions. Overlap fraction is defined as the proportion of ASVs in the coalesced community that were present in both parental communities before mixing. Nutr-: mean = 0.10 ± 0.10 (n=90); Base: mean = 0.24 ± 0.26 (n=83); Nutr+: mean = 0.18 ± 0.25 (n=90). The relatively low overlap fractions across all conditions indicate that most surviving ASVs in coalesced communities were detected in only one of the two parental communities before mixing.

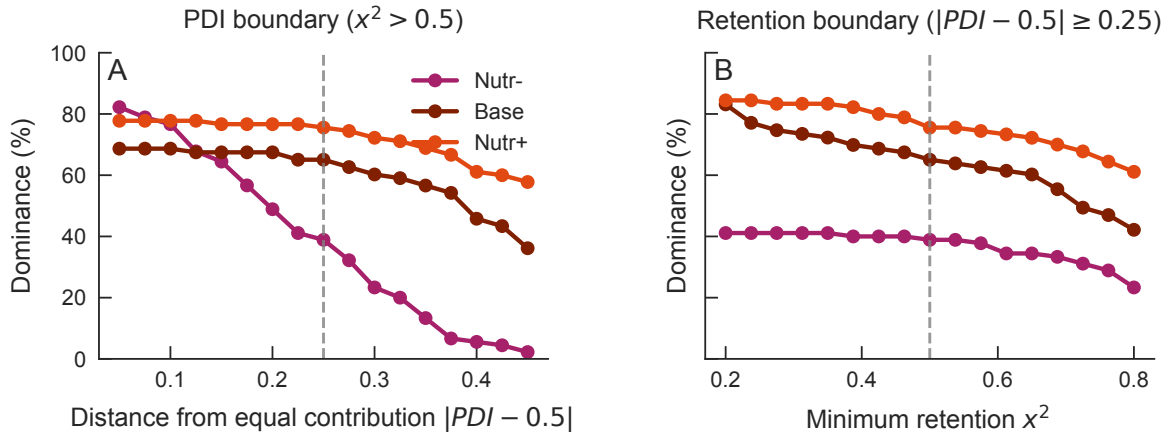


Supplementary Fig. 35. Fraction of coalesced ASVs present in both parental communities for natural sample-derived communities. Distribution of overlap fraction across coalescence events in three nutrient conditions. Overlap fraction is defined as the proportion of ASVs in the coalesced community that were present in both parental communities before mixing. Nutr-: mean = 0.13 ± 0.09 (n=30); Base: mean = 0.09 ± 0.13 (n=30); Nutr+: mean = 0.23 ± 0.15 (n=30). Similar to synthetic communities (Supplementary Fig. 34), the relatively low overlap fractions indicate that most surviving ASVs in coalesced communities were detected in only one of the two parental communities before mixing.

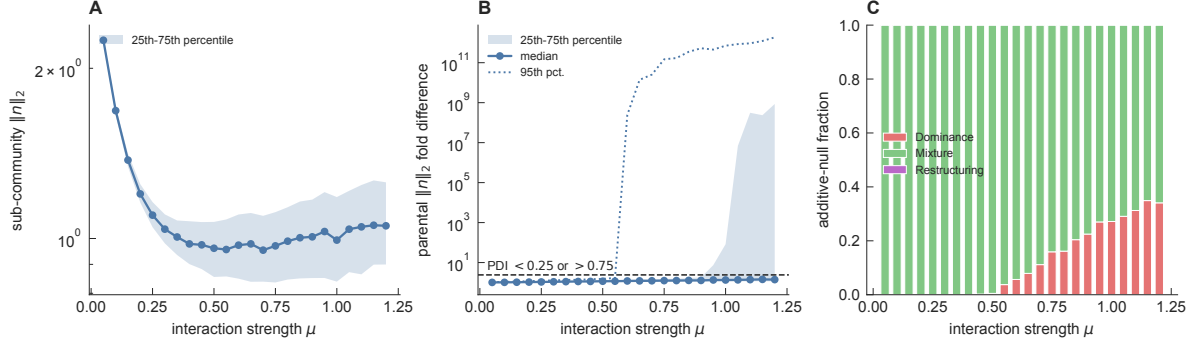
Shared-fate, classification-boundary, and additive-null controls



Supplementary Fig. 36. Excess same-parent shared fate scales with interaction-strength parameter μ . This auxiliary analysis focuses on the within-parent component of the same-vs-cross pairwise selection correlation metric shown in Fig. 2D, using the fate-concordance implementation defined in Supplementary Methods and applied in Supplementary Note 7. Across the full simulated μ sweep ($\mu = 0.05\text{--}1.20$), excess same-parent shared fate (observed same-parent shared fate minus a parental-affiliation-shuffled null that randomizes parental-affiliation labels while preserving the coalesced community composition) tracks the interaction-strength parameter with Pearson $r = 0.870$, $p = 3.2 \times 10^{-8}$. Because the null holds composition fixed and only permutes labels, the observed enrichment is not expected from parental-affiliation label structure alone and is consistent with interaction-mediated correlated species fates during coalescence.

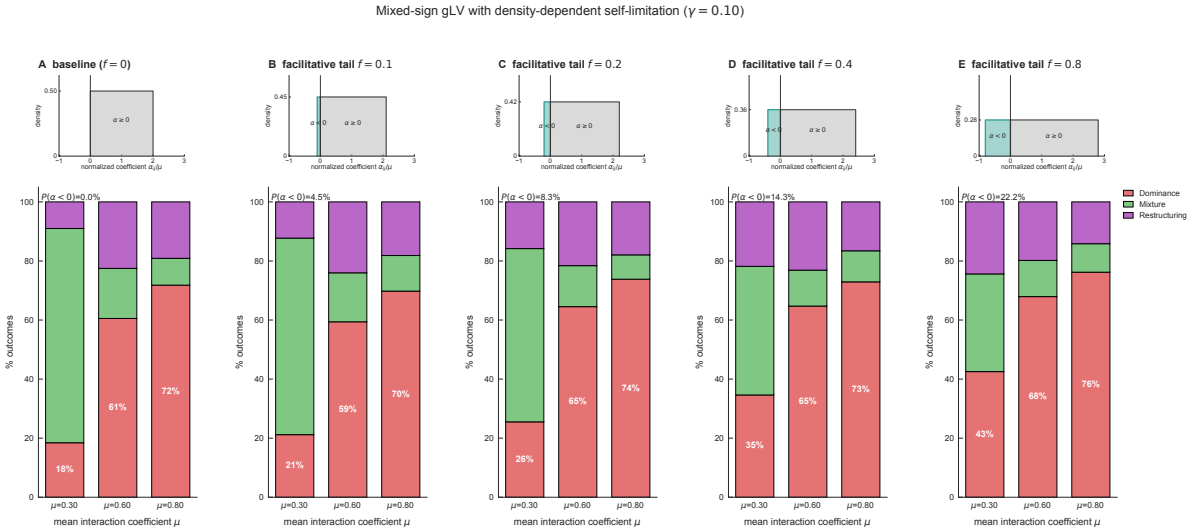


Supplementary Fig. 37. Boundary-sensitivity analysis for the Dominance classification. **(A)** Dominance fractions after varying the PDI boundary while holding the retention boundary at $r^2 > 0.5$. **(B)** Dominance fractions after varying the retention boundary while holding the PDI boundary at the manuscript baseline. Dashed lines mark the manuscript baseline. This sensitivity analysis is descriptive.

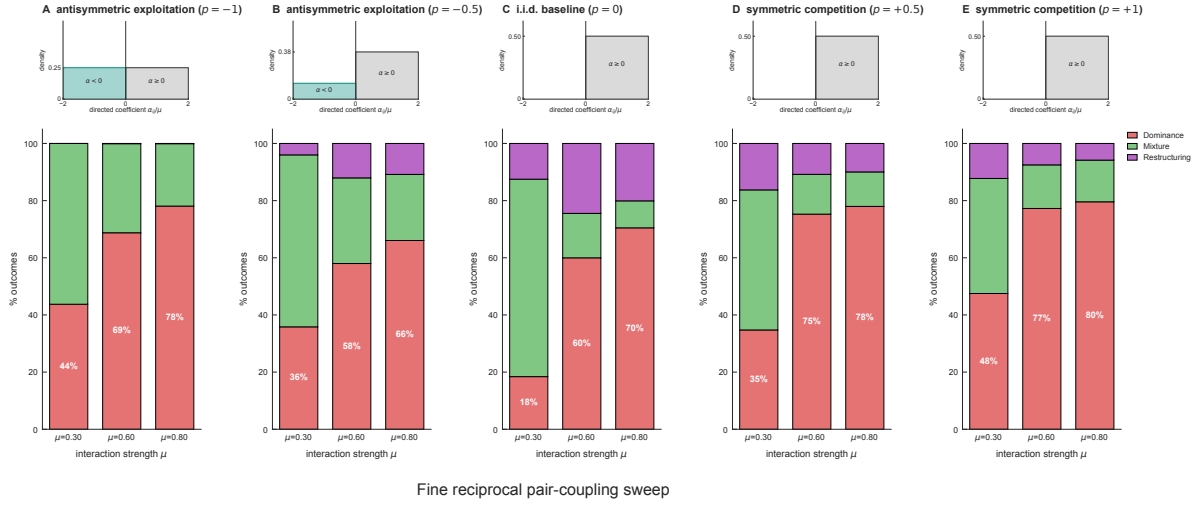


Supplementary Fig. 38. Simulation parental-community norm imbalance and simple additive null model classifications across interaction-strength parameter values. **(a)** Euclidean norms of assembled parental-community abundance vectors across μ ; line shows the median and shading shows the 25th–75th percentile range. **(b)** Fold difference in Euclidean norm between the two parental communities for each simulated coalescence event; line shows the median and shading shows the 25th–75th percentile range, and dotted line shows the 95th percentile. The dashed horizontal line marks the fold-difference threshold corresponding to the Dominance boundary for the simple additive null model. **(c)** Classification fractions for the simple additive null model computed from the same paired parental-community vectors at each μ .

Mixed-sign and coefficient-structure simulation robustness

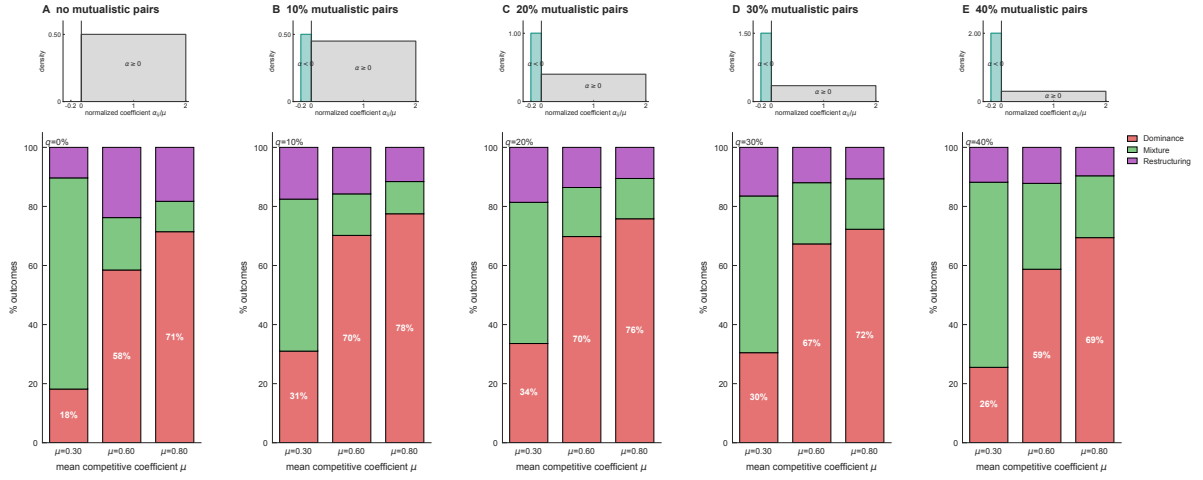


Supplementary Fig. 39. Mixed-sign facilitative-tail gLV sensitivity analysis. Off-diagonal coefficients are drawn iid from $\alpha_{ij} \sim U[-f\mu, (2 + f)\mu]$, preserving $E[\alpha_{ij}] = \mu$ while increasing the expected facilitative fraction $P(\alpha_{ij} < 0) = f/(2 + 2f)$ from 0 to 22.2% at $f = 0.80$. Negative α_{ij} values correspond to positive ecological interactions under the sign convention used in the gLV equation. Top insets show the labeled analytic sampling density of α_{ij}/μ . Teal indicates negative/facilitative support and gray indicates nonnegative support. Dynamics include a stabilizing density-dependent self-limitation term, $\dot{n}_i = n_i(1 - (An)_i - 0.1n_i^2)$. Each panel fixes one facilitative-tail strength f and shows Dominance / Mixture / Restructuring fractions at $\mu \in \{0.30, 0.60, 0.80\}$, with 200 pools and 6 coalescence events per cell.

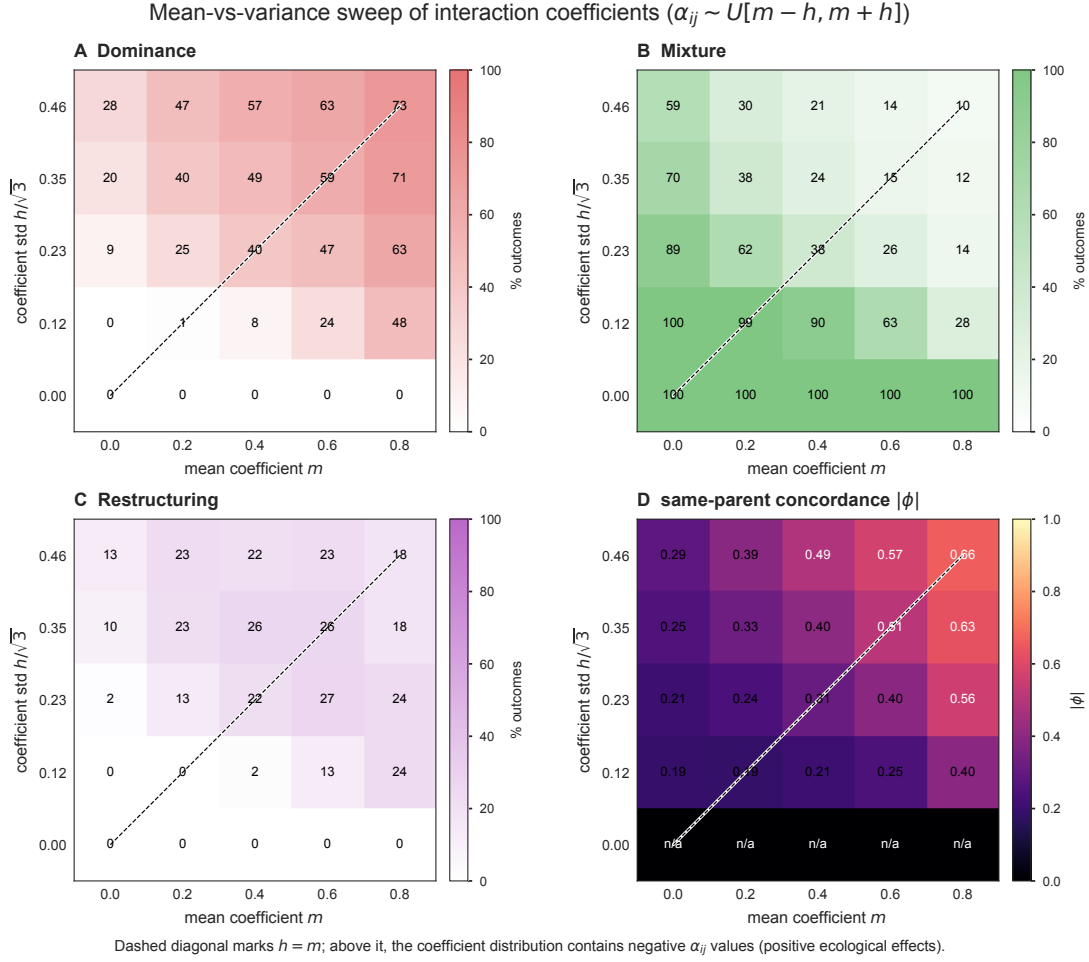


Supplementary Fig. 40. Reciprocal pair-coupling sensitivity analysis. In this extension, the reciprocal pair-coupling structure between coefficients α_{ij} and α_{ji} is varied. **Top:** Coarse sweep ordered from an antisymmetric exploitation structure ($p < 0$, one coefficient in a converted pair is negative) through the iid competitive baseline ($p = 0$) to a symmetric competition structure ($p > 0$, both converted coefficients are positive and equal). Each panel fixes one p and shows Dominance / Mixture / Restructuring fractions at $\mu \in \{0.30, 0.60, 0.80\}$. Top insets show the labeled analytic directed-coefficient distribution implied by each p value. **Bottom:** Fine sweep with p from -1 to $+1$ in increments of 0.2 . Each panel fixes μ and shows stacked outcome fractions across p . The black line marks the Dominance fraction. This pair-coupling extension shows that the μ -dependent Dominance trend is not restricted to iid reciprocal coefficients.

Weak mutualistic-pair sweep ($\alpha_{ij}, \alpha_{ji} \sim U[-0.2\mu, 0]$; $\gamma = 0.10$)

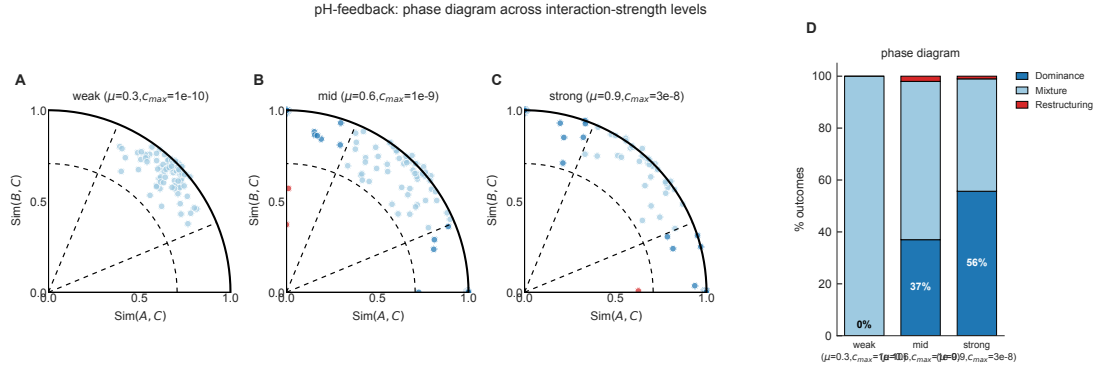


Supplementary Fig. 41. Weak mutualistic-pair fraction sensitivity analysis. For a controlled fraction $q \in \{0, 0.10, 0.20, 0.30, 0.40\}$ of unordered species pairs, both reciprocal coefficients are sampled from $\alpha_{ij}, \alpha_{ji} \sim U[-0.2\mu, 0]$, corresponding to weak positive ecological interactions under our sign convention. All remaining off-diagonal coefficients are sampled from the baseline competitive distribution $U[0, 2\mu]$. Dynamics include the same stabilizing density-dependent self-limitation term used in the facilitative-tail analysis, $\dot{n}_i = n_i(1 - (An)_i - 0.1n_i^2)$. Each panel fixes one mutualistic-pair fraction q and shows Dominance / Mixture / Restructuring fractions at $\mu \in \{0.30, 0.60, 0.80\}$, with 200 pools and 6 coalescence events per cell. Weak mutualistic pairs did not remove the μ -dependent Dominance trend.

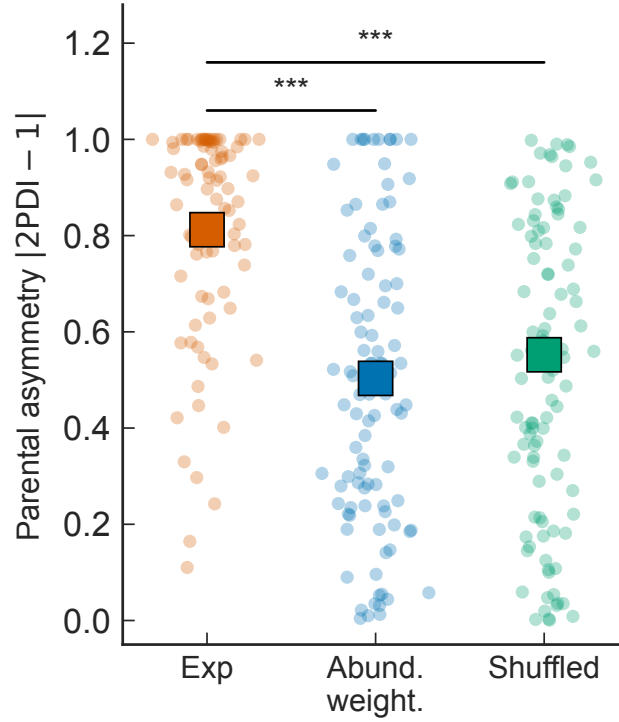


Supplementary Fig. 42. Mean and variance of the interaction-coefficient distribution. Off-diagonal coefficients are sampled from $\alpha_{ij} \sim U[m - h, m + h]$, with mean coefficient $m \in \{0, 0.2, 0.4, 0.6, 0.8\}$ and standard deviation $h/\sqrt{3}$ for $h \in \{0, 0.2, 0.4, 0.6, 0.8\}$. Negative α_{ij} values, corresponding to positive ecological interactions under our sign convention, occur above the dashed $h = m$ line. Dynamics include the same stabilizing density-dependent self-limitation term used in the facilitative-tail analysis, $\dot{n}_i = n_i(1 - (An)_i - 0.1n_i^2)$. Heatmaps show Dominance, Mixture, Restructuring, and same-parent fate concordance $|\phi|$ across the 5×5 grid, where $|\phi|$ denotes the absolute excess same-parent shared-fate signal relative to an affiliation-shuffled null. Each cell uses 200 pools and 6 coalescence events per pool. Both average inhibitory effect and coefficient heterogeneity contribute to the Dominance regime in this mean-versus-variance sweep.

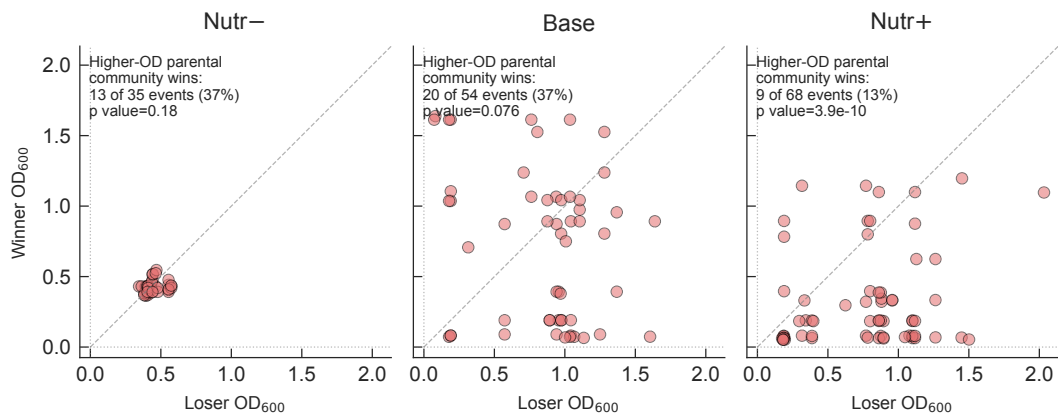
Alternative-mechanism and passive-null controls



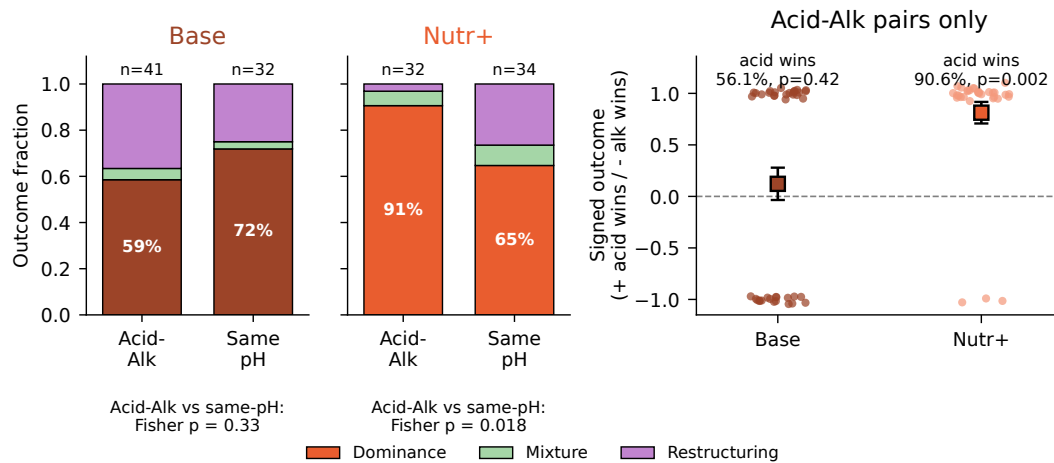
Supplementary Fig. 43. Ratzke–Gore-style pH-feedback alternative model. A pH-feedback model motivated by pH-mediated microbial interactions can generate Dominance-like asymmetric outcomes over an intermediate feedback range. In this implementation, Restructuring was rare across feedback strengths ($\sim 1\%$; 3/297 simulated events), whereas Restructuring is more common in the effective-interaction gLV framework ($\sim 22\%$ across μ values). The response shape differs from the monotonic gLV interaction-strength transition, so pH feedback is best interpreted as a contributing mechanism rather than a complete alternative explanation for the full nutrient-dependent Dominance trend.



Supplementary Fig. 44. Abundance-skew null models do not explain the observed parental asymmetry in Base-medium coalescence outcomes. Direction-independent parental asymmetry, $|2PDI - 1|$, is compared between experimental Base-medium coalescence outcomes and two null models: abundance-weighted random selection, in which survival probability is proportional to abundance in the combined parental-community pool, and shuffled abundance, in which abundances are randomly permuted among species within each parental community before neutral mixing. Points show sampled outcomes and squares show group means. Both null models produced significantly lower parental asymmetry than the experimental data (Mann–Whitney U tests; *** indicates $p < 0.001$).



Supplementary Fig. 45. Higher parental-community OD_{600} does not determine Dominance direction. For Dominance-class coalescence events in Nutr–, Base, and Nutr+, winner OD_{600} is plotted against loser OD_{600} . The dashed line marks equal winner and loser OD ; points above this line would indicate cases where the higher- OD parental community won. Text in each panel reports the number and fraction of Dominance events in which the higher- OD parental community won, with two-sided binomial tests against a 50% expectation.



Supplementary Fig. 46. Strict parental-community pH contrast is most informative in Nutr+. Base and Nutr+ panels show stacked Dominance, Mixture, and Restructuring fractions for strict acidic–alkaline and strict same-pH pairings. Parental communities were classified as acidic when $\text{pH} < 6.5$ and alkaline when $\text{pH} > 7.5$; intermediate-pH pairings were excluded. Statistical tests compare Dominance with pooled Mixture and Restructuring outcomes. Strict acidic–alkaline pairings were enriched for Dominance in Nutr+ (29/32 versus 22/34 events; OR = 5.27, Fisher $p = 0.018$), but not in Base medium (24/41 versus 23/32 events; OR = 0.55, Fisher $p = 0.33$). The Acid-Alk pairs-only panel shows winner direction within strict acidic–alkaline pairings: the acidic parental community dominated 29/32 Nutr+ events (90.6%), compared with 23/41 Base events (56.1%; Extended Data Fig. 7). Dots show individual binary winner-direction outcomes; squares show mean $\pm$ s.e.m.

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